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# TRANSFER AND ORBIT OPTIMIZATION FOR L4 MISSIONS

*END OF STUDY INTERNSHIP*

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REMI CARESCHE  
ISAE-SUPAERO TAS ASTRO  
NASA Goddard Space Flight Center  
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NASA GSFC Mentor: Dr. Ariadna Farrés Basiana  
ISAE SUPAERO Mentor: Prof. Stéphanie Lizy-Destrez



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# 1 Acknowledgment

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I also wish to thank my entire team in the Navigation and Mission Design Branch for welcoming me and for their willingness to share their knowledge. I would also like to acknowledge the other interns and researchers I met at GSFC, whose enthusiasm made this internship even more enjoyable.

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## 2 Abstract

This work presents the design and optimization of transfer trajectories between libration point orbits in the Sun/Earth system, with a particular focus on the triangular Lagrange point  $L_4$ . The study was carried out during an internship within the Navigation and Mission Design Branch (Code 595) at the NASA Goddard Space Flight Center, and addresses the challenge of transferring a spacecraft from a periodic Lyapunov orbit around  $L_1$  to a periodic Lyapunov orbit around  $L_4$ .

The analysis relies on the Circular Restricted Three-Body Problem (CRTBP) as a dynamical framework to model the motion of a spacecraft under the combined gravitational influence of the Sun and the Earth. Periodic orbits around the libration points are computed using a Newton-type shooting method, complemented by modal initialization near  $L_4$  to account for the absence of publicly available reference solutions. Transfer trajectories are then designed by exploiting the unstable manifold of the  $L_1$  orbit and connecting it to the vicinity of the  $L_4$  orbit through a two-impulse maneuver strategy.

The first phase was implemented in Python, providing analytical understanding of the dynamical structures. The second phase employed the General Mission Analysis Tool (GMAT) to validate the transfers within a high-fidelity dynamical environment, including ephemerides and third-body perturbations. A dual approach was adopted to ensure both theoretical insight and operational relevance.

The results confirm the feasibility of low-thrust-like, manifold-assisted transfers toward the Sun/Earth  $L_4$  region using impulsive  $\Delta V$  maneuvers. The study highlights the complementarity between dynamical systems theory and numerical mission design, contributing to future heliophysics missions operating at the triangular Lagrange points.

### 3 Introduction

The exploration of libration point dynamics has become a cornerstone of modern astrodynamics, enabling spacecraft to operate in regions where gravitational and centrifugal forces balance. Missions such as SOHO or JWST have demonstrated the scientific and operational advantages of these locations, offering stable thermal environments, continuous observation geometries, and efficient transfer opportunities.

In this context, the Sun/Earth triangular Lagrange points ( $L_4$  and  $L_5$ ) have become points of interest for heliophysics and space-weather applications. Positioned  $60^\circ$  ahead and behind Earth, these points provide complementary perspectives on solar activity and interplanetary space. For instance, a spacecraft located at  $L_5$  can continuously observe active regions on the Sun before they rotate into Earth's view, enabling early detection of solar eruptions and coronal mass ejections that could impact satellites and communication systems. Severe space-weather events have the potential to cause global economic losses estimated at up to \$2 trillion in a worst-case scenario [Council, 2008], underscoring the importance of continuous monitoring from multiple vantage points such as  $L_4$  and  $L_5$ .

The project was conducted within the Navigation and Mission Design Branch (Code 595) at the NASA Goddard Space Flight Center. It focuses on the mission design problem associated with reaching and operating near the Sun/Earth  $L_4$  point. Specifically, it aims to investigate transfer trajectories between a periodic Lyapunov orbit around  $L_1$  and a periodic Lyapunov orbit around  $L_4$ , under the dynamical framework of the Circular Restricted Three-Body Problem (CRTBP).

The report first presents the institutional and organizational context of the Goddard Space Flight Center, followed by a detailed description of the theoretical background of the CRTBP and the properties of the Lagrange points. Subsequent sections discuss the computation of periodic orbits around the libration points using Newton's method and modal initialization, and finally, the design and optimization of transfer trajectories linking  $L_1$  and  $L_4$  through manifold-based strategies.

## 4 Goddard Space Flight Center

The NASA Goddard Space Flight Center (GSFC) is a research and development facility located in Greenbelt, Maryland, USA. Established on May 1, 1959, GSFC was NASA's first space flight complex and was named in honor of Dr. Robert H. Goddard (1882–1945), one of the pioneers of modern rocketry [NASA GSFC, a].

The center's mission spans the entire lifecycle of space science projects, from conceptual design and engineering to launch operations and on-orbit mission support. It contains the largest community of scientists, engineers, and technologists within NASA (around 10,000 people), all contributing to the study of Earth, the Sun, the solar system, and the universe.

The center is also responsible for key operational tasks, including the management of the Hubble Space Telescope and communications with the International Space Station through NASA's Tracking and Data Relay Satellite System and the Deep Space Network.

### 4.1 Historical Background

The creation of the center was closely linked to the early days of NASA and the growing need for a dedicated center to develop and operate scientific satellites. Before the foundation of the agency in 1958, the "Vanguard Project", which launched one of the first U.S. satellites, had already highlighted the importance of a facility specializing in scientific payloads and satellite operations [NASA GSFC, a].

Following its early contributions to spaceflight infrastructure, the Center rapidly grew into a key actor in scientific research. Among its major contributions, a few missions stand out for their scientific impact and long-term legacy [Launius, 1994]:

- **Landsat Program (1972–present):** In collaboration with the U.S. Geological Survey, GSFC has operated the Landsat satellite series for over 50 years. This program provides consistent, global, multispectral imagery of Earth's surface and has become an essential tool in environmental monitoring, agriculture, urban development, and climate science.

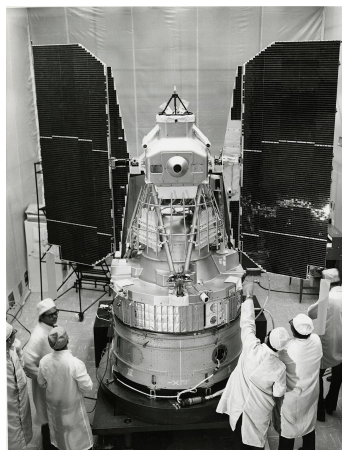


Figure 1: Landsat 1 inside the clean room  
Source: Landsat 1



- **Hubble Space Telescope (1990–present):** Hubble is one of the most influential space observatories ever launched. It has revolutionized modern astrophysics by providing deep-field images (images of distant regions of the universe), measuring the expansion of the universe, and characterizing distant galaxies and exoplanetary systems.



Figure 2: Hubble Space Telescope during STS-82 mission  
*Source: STS-82*

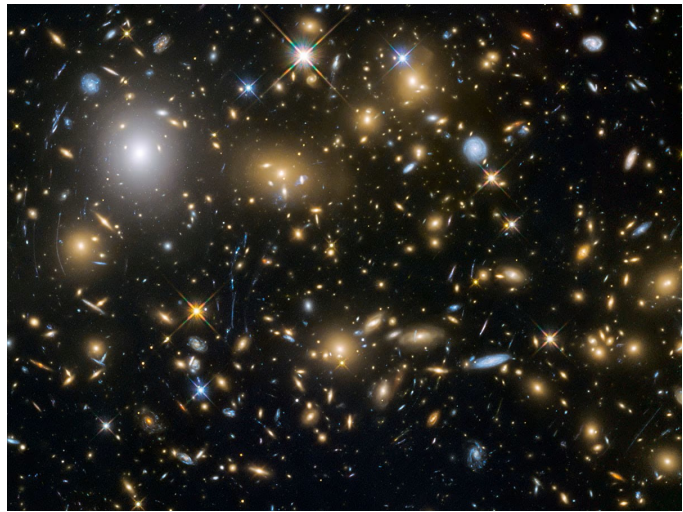


Figure 3: Hubble picture of a galaxy cluster  
*Source: The Atlantic*

- **James Webb Space Telescope (JWST, 2021–present):** Deployed at the Sun/Earth L2 point, the James Webb Space Telescope enables infrared and visible observations of the early universe, star formation regions, and exoplanet atmospheres, extending and complementing the legacy of Hubble.



Figure 4: James Webb picture of a galaxy cluster  
*Source: Science News*

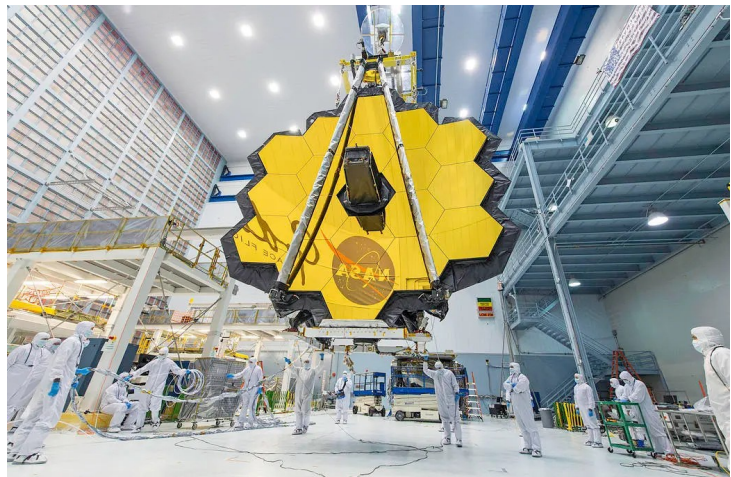


Figure 5: James Webb Space Telescope inside GSFC clean room  
*Source: NASA [nasa.gov](https://www.nasa.gov)*

The Goddard Space Flight Center has long served as a central pillar in NASA's scientific and operational architecture. Its role has extended far beyond individual missions,

contributing fundamentally to the agency’s capacity to plan, operate, and support space exploration at scale. Historically, GSFC played a vital part in the success of the Apollo program by developing and managing tracking and communication systems, and later provided key infrastructure during the Space Shuttle era.

## 4.2 Organizational Structure

The NASA Goddard Space Flight Center is organized into eight directorates, each responsible for different aspects of scientific, technical, and operational activities [NASA GSFC, a]. Among them, the Engineering and Technology Directorate (Code 500) plays a central role in the design and development of spacecraft systems, instruments, and mission architecture. This directorate encompasses a wide range of activities, from systems engineering and spacecraft integration to trajectory design and flight dynamics.

Within this directorate, the Navigation and Mission Design Branch (Code 595) focuses on astrodynamics and mission analysis. The team specializes in transfer trajectory design, orbit determination, and station-keeping strategies across various orbital regimes, including low Earth orbit, geostationary orbit, and libration point orbits. It supports both robotic and human spaceflight programs by evaluating mission feasibility,  $\Delta V$  requirements, and launch opportunities. The branch has contributed to several flagship missions, including:

- **James Webb Space Telescope** : Design of the Sun/Earth L2 transfer and orbit maintenance strategies.
- **Nancy Grace Roman Space Telescope**: Trajectory optimization and operational planning for L2 insertion.
- **Lunar Gateway**: Navigation planning and orbit design for cis-lunar operations under the Artemis program.
- **OSIRIS-REx**: Support in navigation and maneuver planning for the asteroid rendez-vous and sample return phases.

In addition to its mission support role, the branch is also the original developer of the open-source software GMAT (General Mission Analysis Tool). This tool has become a widely used reference in astrodynamics and mission design, supporting trajectory optimization, orbit propagation, and  $\Delta V$  budgeting. In parallel, the branch also maintains a close operational interface with the Flight Dynamics Facility (FDF), which serves as the real-time control room for flight dynamics operations at GSFC, ensuring the continuity between mission planning and execution. As part of my internship, I have been integrated into the Navigation and Mission Design Branch and involved in mission design tasks.

## 4.3 Future Directions

Looking toward the near future, GSFC is deeply involved in final integration and testing phases of the Nancy Grace Roman Space Telescope [NASA GSFC, b], a major observatory poised to complement Hubble and James Webb Space telescope. Beyond that, the center is preparing to support upcoming Lunar Gateway/Artemis missions, which will



extend human and robotic operations to lunar orbit.

In the current strategic and economic context, both NASA and the Goddard Space Flight Center are facing significant challenges that are shaping the future of space exploration. Budgetary constraints and evolving political priorities have led to growing internal tensions across the agency, with increasing pressure to reconcile short-term cost control with the long-term ambition of deep space missions. These tensions have already resulted in delays, scope reductions, and even the cancellation of major programs such as OSAM-1, a flagship satellite servicing mission. These issues highlight the difficulty of maintaining a stable innovation pipeline under uncertain financial and institutional conditions.

The center is also undergoing a profound internal transition through the Deferred Resignation Program (DRP) [Smith, 2025], which has led to a wave of voluntary departures among civil servants and contractors. GSFC has been particularly impacted, with over 600 positions affected and close to 1,000 employees expected to leave. This loss of personnel not only raises concerns about knowledge transfer and workforce continuity, but also adds strain on the teams that remain, especially in mission-critical areas such as flight dynamics and systems engineering. In spite of these structural challenges, the center remains committed to delivering on its current missions.

## 5 Project context

Before entering into the technical details of the mission design, it is useful to summarize the objectives of this study. The project aims to investigate and develop transfer trajectories connecting two equilibrium points of the Sun/Earth system, namely  $L_1$  and  $L_4$ . The transfers are to be realized under the assumption of a chemical propulsion system, modeled as impulsive  $\Delta V$  maneuvers. This choice reflects both the heritage of many operational missions and the need for a simplified but realistic framework in preliminary design.

The analysis will be carried out using a dual approach: Python-based tools will be employed to implement the Circular Restricted Three-Body Problem. The solutions obtained in the CRTBP will be exported and validated in GMAT, which provides a higher-fidelity dynamical environment, including ephemerides, perturbations, and maneuver modeling. This combination ensures that the methodology is both grounded in analytical dynamical systems theory and consistent with operational mission design practice at GSFC. In this framework, Python serves as a simple environment for implementing the CRTBP, exploring dynamical structures, and prototyping transfer strategies. GMAT, on the other hand, provides the necessary high-fidelity validation tool that is not only widely adopted in the research community but also formally qualified for NASA mission analysis.

### 5.1 Circular Restricted Three-Body Problem

The Circular Restricted Three-Body Problem (CRTBP) is a classical dynamical model designed to study the motion of a massless spacecraft under the gravitational influence of two massive bodies, called the primaries. The model is based on three main assumptions:

- The spacecraft has negligible mass compared to the primaries and does not perturb their motion.
- The two primaries (here the Sun and the Earth) revolve on perfectly circular orbits around their barycenter.
- The motion of the spacecraft is governed only by the Newtonian gravitational attraction of the two primaries, expressed in a uniformly rotating reference frame.

These simplifications remove the complexity of the full  $N$ -body problem while preserving the essential dynamical structures of equilibrium point, periodic orbit, ... The CRTBP is therefore a natural starting point for trajectory design near libration points, where the gravitational effects of both primaries must be accounted for simultaneously [Koon et al., 2022].

#### 5.1.1 Rotating Reference Frame

To formulate the equations, a synodic (rotating) frame is introduced, rotating with angular velocity equal to the mean motion of the primaries. In this frame, the primaries are

fixed on the  $x$ -axis, which makes it convenient to study equilibrium and relative motion. A classical non-dimensionalization is adopted:

$$a = 1, \quad n = 1 \quad (\text{mean motion}), \quad G(m_1 + m_2) = 1 \quad (1)$$

where  $a$  is the distance between the primaries,  $n$  their angular velocity, and  $G$  the gravitational constant. The unit system implies that one length unit corresponds to 1 AU, one time unit to  $\approx 58$  days, and one velocity unit to  $\approx 29.8$  km/s in the Sun/Earth case. The mass parameter is defined as:

$$\mu = \frac{m_2}{m_1 + m_2}, \quad 0 < \mu < 1 \quad (2)$$

which, for the Sun/Earth system, takes the very small value  $\mu \approx 3.054200 \times 10^{-6}$ . In this frame, the primaries are located at:

$$\mathbf{r}_{m_1} = (-\mu, 0, 0), \quad \mathbf{r}_{m_2} = (1 - \mu, 0, 0) \quad (3)$$

### 5.1.2 Equations of Motion

Let  $\mathbf{r} = (x, y, z)$  be the spacecraft position. Its distances to the primaries are:

$$r_1 = \sqrt{(x + \mu)^2 + y^2 + z^2}, \quad r_2 = \sqrt{(x - 1 + \mu)^2 + y^2 + z^2} \quad (4)$$

The effective potential in the rotating frame is given by:

$$\Omega(x, y, z) = \frac{1 - \mu}{r_1} + \frac{\mu}{r_2} + \frac{1}{2}(x^2 + y^2) \quad (5)$$

The equations of motion of the spacecraft in space are then written as [Singh et al., 2021]:

$$\ddot{x} - 2\dot{y} = \frac{\partial \Omega}{\partial x} = x - (1 - \mu)\frac{x + \mu}{r_1^3} - \mu\frac{x - 1 + \mu}{r_2^3} \quad (6)$$

$$\ddot{y} + 2\dot{x} = \frac{\partial \Omega}{\partial y} = y - (1 - \mu)\frac{y}{r_1^3} - \mu\frac{y}{r_2^3} \quad (7)$$

$$\ddot{z} = \frac{\partial \Omega}{\partial z} = -(1 - \mu)\frac{z}{r_1^3} - \mu\frac{z}{r_2^3} \quad (8)$$

Note that, the  $1/r$  terms are Newtonian gravity from the primaries.  $\frac{1}{2}(x^2 + y^2)$  yields the centrifugal contribution. The  $\pm 2$  couplings are Coriolis accelerations.

## 5.2 Lagrange Points and Hill Regions

In the CRTBP, equilibrium solutions correspond to positions where the net acceleration and velocity in the rotating frame vanishes. This requires

$$\ddot{x} = \ddot{y} = \ddot{z} = 0, \quad \dot{x} = \dot{y} = \dot{z} = 0 \quad (9)$$

which is equivalent to solving

$$\nabla \Omega(x, y, z) = 0 \quad (10)$$

Explicitly :

$$\frac{\partial \Omega}{\partial x} = x - (1 - \mu) \frac{x + \mu}{r_1^3} - \mu \frac{x - 1 + \mu}{r_2^3} = 0 \quad (11)$$

$$\frac{\partial \Omega}{\partial y} = y \left( 1 - \frac{1 - \mu}{r_1^3} - \frac{\mu}{r_2^3} \right) = 0 \quad (12)$$

$$\frac{\partial \Omega}{\partial z} = -z \left( \frac{1 - \mu}{r_1^3} + \frac{\mu}{r_2^3} \right) = 0 \quad (13)$$

### 5.2.1 The Five Lagrange Points

The system admits five equilibrium solutions, called the Lagrange points (figure 6).

- Collinear points  $L_1, L_2, L_3$ : These lie on the  $x$ -axis ( $y = z = 0$ ) and are obtained by solving a nonlinear equation in  $x$  [Koon et al., 2022]:

$$x - (1 - \mu) \frac{x + \mu}{|x + \mu|^3} - \mu \frac{x - 1 + \mu}{|x - 1 + \mu|^3} = 0 \quad (14)$$

In the Sun/Earth system,  $L_1$  and  $L_2$  are located about  $1.5 \times 10^6$  km from Earth. These points are linearly unstable, and their dynamics are governed by the presence of stable, center (manifold containing periodic and quasi-periodic orbits) and unstable invariant manifolds. Such manifolds structure the dynamic around the equilibrium point and are widely exploited in mission design for low- $\Delta V$  transfers.

- Triangular points  $L_4, L_5$ : These form equilateral triangles with the primaries:

$$(x, y, z) = \left( \frac{1}{2} - \mu, \pm \frac{\sqrt{3}}{2}, 0 \right) \quad (15)$$

For the Sun/Earth system, they are located  $60^\circ$  ahead and behind Earth on its orbit. Unlike the collinear points,  $L_4$  and  $L_5$  are linearly stable in the Sun/Earth case, allowing for long-lived libration orbits.

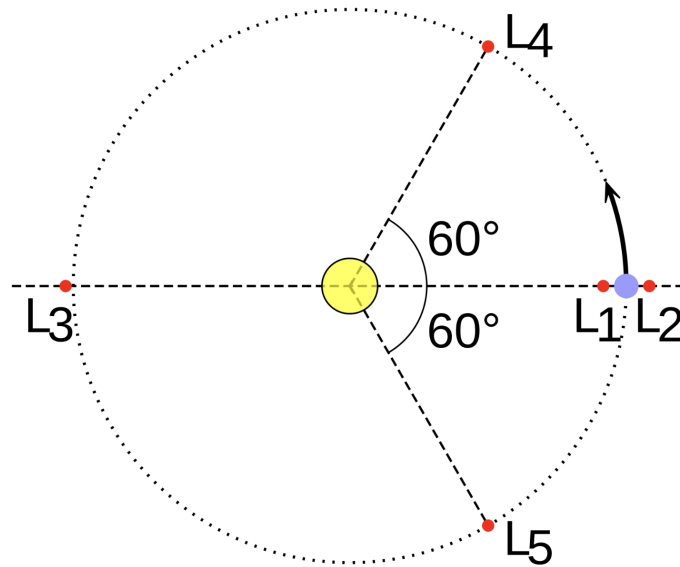


Figure 6: Lagrange points in Sun/Earth system  
Source: Lagrange Point

At the Lagrange points, gravitational attraction from the primaries and the inertial (centrifugal) acceleration balance perfectly in the rotating frame. Collinear points provide gateways for transit between dynamical regions, while triangular points correspond to corotating configurations.

### 5.2.2 Hill Regions

The Jacobi constant is defined as :

$$C = 2\Omega(x, y, z) - (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) \quad (16)$$

It is conserved along any unforced trajectory in the CRTBP and provides a global integral of motion for the system. When the spacecraft velocity vanishes, the corresponding surface  $2\Omega(x, y, z) = C$  defines the so-called zero-velocity surface, which acts as a boundary between dynamically forbidden and admissible regions. The interior of this surface, where  $2\Omega(x, y, z) \geq C$ , is called the Hill region. Within the Hill region the motion is kinematically possible, whereas outside it is forbidden for trajectories with the given Jacobi constant [Koon et al., 2022].

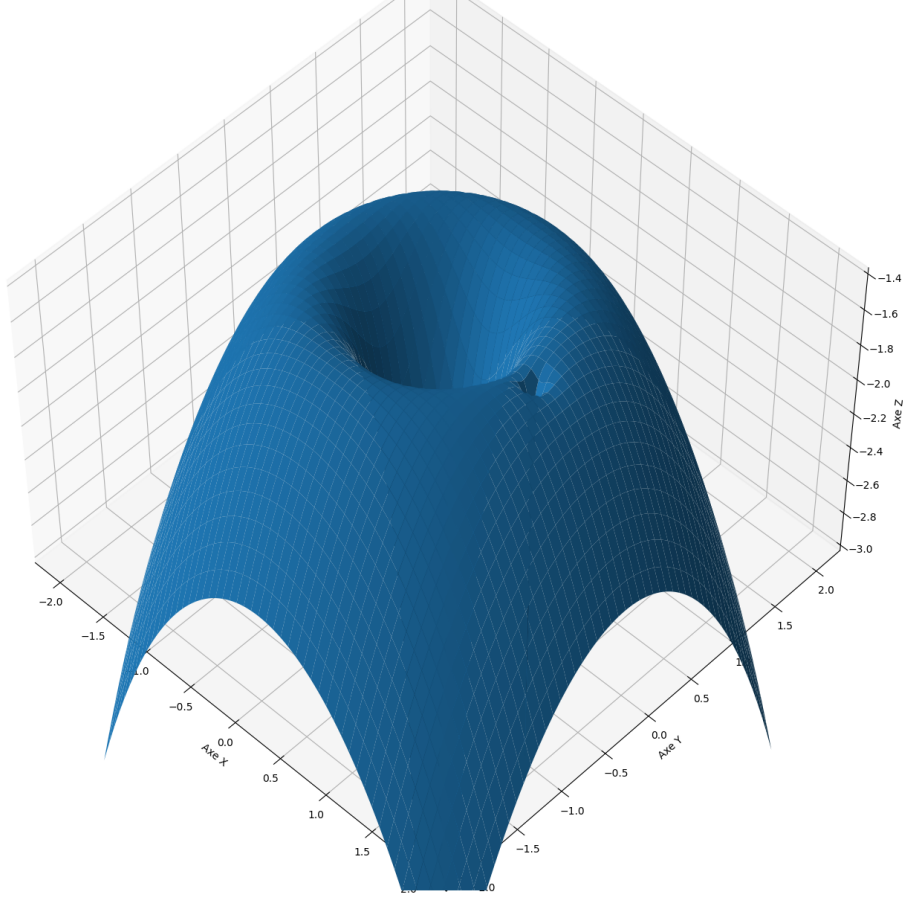


Figure 7: Boundary of the Hill region

For large  $C$  (low kinetic energy), the Hill region is disconnected and the spacecraft is trapped near one primary. As  $C$  decreases, new paths open near  $L_1$  and  $L_2$ , enabling transit trajectories between interior and exterior regions. This energy-based view is central to understanding the feasibility of inter-libration transfers. This pattern is shown in figure 8.



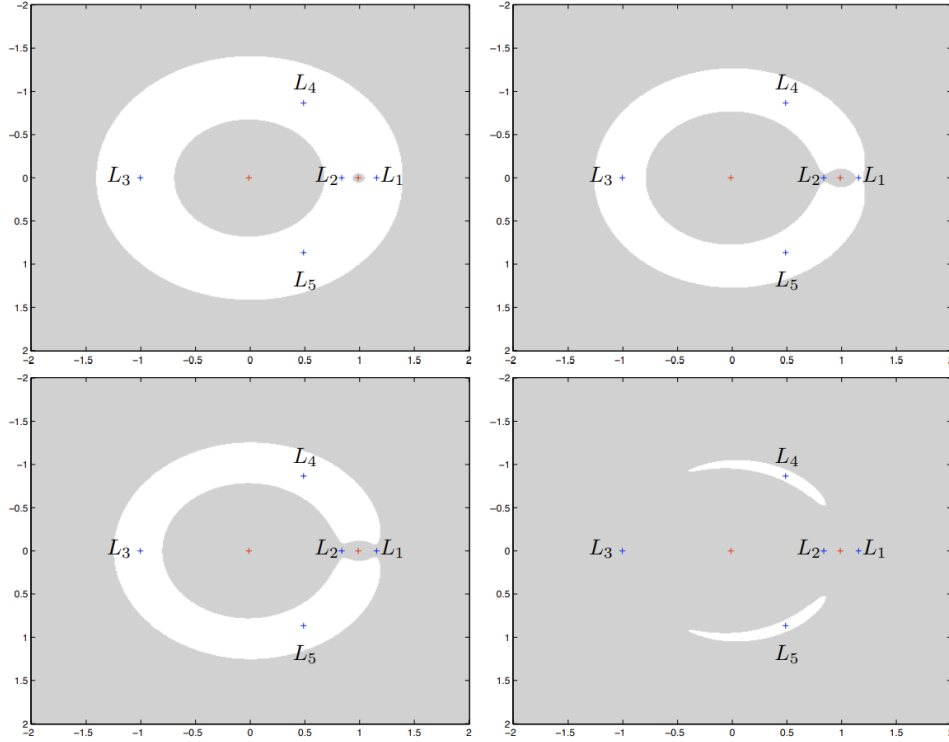


Figure 8: Inter-libration transfers (Hill region) [Daoud, 2011], each plot is a different value of  $C$  and possible motion is in gray

### 5.3 Scientific Context of an L4 Mission

The Sun/Earth  $L_4$  point offers a unique position for heliophysics. Located  $60^\circ$  ahead of Earth in its orbit,  $L_4$  observes regions of the solar surface that have just rotated out of Earth's view. While this does not provide early-warning capability, it enables continuous monitoring of active regions after their transit across the Earth-facing hemisphere. Such observations are valuable for studying the persistence and recurrence of sunspots, active regions, and coronal holes over multiple solar rotations [Sood and Howell, 2019].

In recent years, growing attention has been given to missions around the triangular Lagrange points  $L_4$  and  $L_5$ , highlighting their strategic role for space-weather monitoring. The European Space Agency, for instance, is preparing the Vigil mission, which will be deployed to  $L_5$  in 2031. From its vantage point  $60^\circ$  behind Earth, Vigil will continuously observe solar regions that are about to rotate into Earth's view, providing an early-warning capability for coronal mass ejections (CMEs) and solar flares. Although Vigil focuses on  $L_5$ , its rationale illustrates the scientific interest of both triangular points: while  $L_5$  supports early-warning forecasts,  $L_4$  provides complementary information, monitoring solar regions after they leave Earth's view.



Figure 9: Vigil mission logo  
Source: ESA Vigil

The present study introduces a spacecraft concept at the Sun/Earth  $L_4$  point, named EOS-L4. The acronym “EOS” refers to the Greek goddess of the dawn, symbolizing the observation of solar activity as it emerges beyond Earth’s horizon. The designation  $L_4$  specifies the triangular Lagrange point  $60^\circ$  ahead of Earth, highlighting the mission’s location. For the purpose of this theoretical investigation, EOS-L4 is assumed to share the baseline design characteristics of Vigil, with a dry mass of 1160 kg and a wet mass of 2500 kg, including about 1300 kg of propellant. These values provide a basis for estimating  $\Delta V$  budgets within the Sun/Earth CRTBP framework. While hypothetical, grounding the concept in an existing mission ensures that the trajectory analysis remain relevant to ongoing developments in heliophysics missions.

## 5.4 Concept of Operations

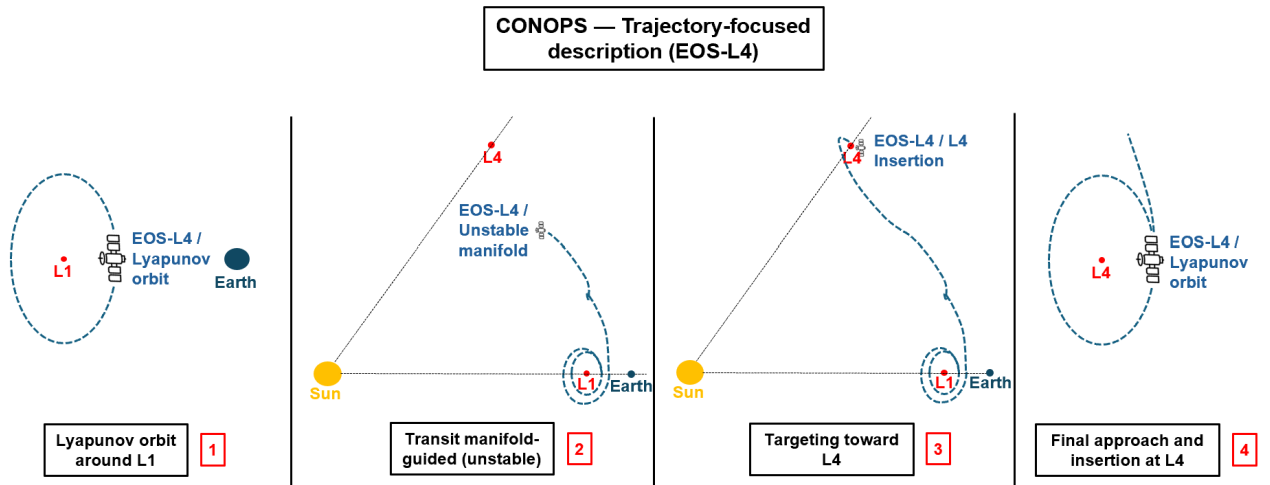


Figure 10: Concept of operations EOS-L4

The Concept of Operations (CONOPS) of EOS-L4, shown in figure 10, focuses on the transfer from a periodic Lyapunov orbit around  $L_1$  to a periodic Lyapunov orbit around  $L_4$  in the Sun/Earth CRTBP system. The initial acquisition of the  $L_1$  Lyapunov orbit (LEOP, Earth-escape, and Lyapunov insertion) is outside the present scope.

1. The spacecraft is established on a Lyapunov orbit around  $L_1$  and phased to the desired departure point. The reference periodic state  $x_{L1}(0)$  and its state-transition matrix (STM) are available for targeting.
2. Let  $v_u$  be the unstable eigenvector of the  $L_1$  Lyapunov orbit at the chosen epoch  $t_0$ . A small perturbation

$$\delta x_0 = \varepsilon v_u \quad (\varepsilon \ll 1) \quad (17)$$

places the spacecraft onto the unstable manifold. In practice, this is realized by applying a very small  $\Delta V_{pert}$  at  $t_0$  to align the state with the manifold direction.

In this study, the  $\Delta V_{pert}$  is negligible compared to the main transfer burns and is thus not included in the budget. This is justified because the manifold rapidly amplifies small unstable displacements, so the needed  $\|\Delta V_{pert}\|$  can be kept at the order of a few m/s.

3. The trajectory follows the unstable manifold from  $L_1$ . On a chosen Poincaré section (e.g., a line at  $30^\circ$  ahead of Earth), a finite burn  $\Delta V_{tgt}$  is executed to match an insertion trajectory to reach Lyapunov orbit around  $L_4$ .
4. As the spacecraft approaches the vicinity of the  $L_4$  periodic orbit around its stable manifold, one insertion burns  $\Delta V_{ins}$  enforce the periodicity constraints and set the desired amplitude/phase of the  $L_4$  Lyapunov.

## 6 Periodic Orbits

Having established the theoretical framework of the CRTBP and the properties of the Lagrange points, the next step is to compute periodic solutions around these equilibrium configurations. These orbits provide natural staging locations for spacecraft and support transfer design through their associated invariant manifolds. Mathematically, a periodic orbit is a trajectory  $x(t)$  in the rotating frame that satisfies:

$$\forall t, \quad x(t + T) = x(t) \quad (18)$$

where  $T$  is the orbit's period. Due to the nonlinearity of the CRTBP, these solutions cannot be expressed in closed form, but can be obtained through numerical continuation and differential correction starting from linearized approximations around the Lagrange points [Yousuf and Kishor, 2022].

Periodic solutions are organized into families:

- Planar Lyapunov orbits: These orbits correspond to one-dimensional oscillations confined to the  $(x, y)$ -plane around the collinear Lagrange points. They originate from the complex conjugate pair of eigenvalues  $(\pm i\omega_1)$  of the linearized dynamics near equilibrium.
- Vertical Lyapunov orbits: These orbits are the same as the previous one but in the  $z$ -direction. They emerge from the second pair of imaginary eigenvalues in the linearized system.
- Halo orbits: Three-dimensional periodic solutions around the collinear points. They arise through continuation of the planar orbits (at some point the planar orbits bifurcate and gives Halo orbits). The result is a three-dimensional orbit that “wraps” around the equilibrium point, forming a bubble structure.
- Lissajous orbits (not developed in this study): Quasi-periodic orbits in the vicinity of libration points, generated by combinations of oscillatory modes.

### 6.1 Computation of Periodic Orbits

The standard approach to compute a periodic orbit is to apply a shooting algorithm combined with a Poincaré section [Chupin, 2016]. A Poincaré section is a lower-dimensional surface, typically defined within the system's phase space, that records the trajectory each time it crosses the surface with a prescribed orientation. This technique reduces the continuous six-dimensional trajectory to a discrete sequence of intersection points, providing a simplified representation of the system's dynamics. The corresponding Poincaré map is the function that associates each crossing to the next one.

An initial guess  $p_0 = x(t_0)$  (dimension 6) is then generated from the linearized dynamics around the equilibrium point, or from existing numerical databases such as those provided by the [Jet Propulsion Laboratory]. The trajectory is propagated forward in time until it intersects the predefined Poincaré section (figure 11), typically chosen as a hyperplane ( $y = 0$  with  $\dot{y} > 0$ ).

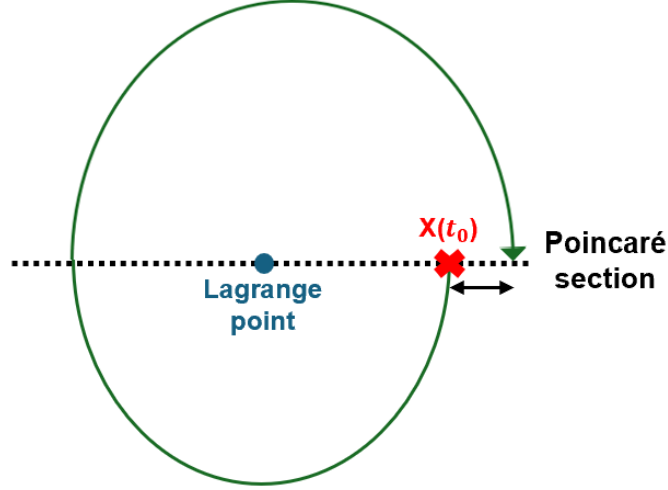


Figure 11: Shooting method

The objective is therefore to start from an initial state  $p_0$  located on the Poincaré section, propagate the trajectory until it intersects the section once, and then continue the propagation until a second crossing is reached. At this point, the following condition must be verified:

$$x(t_f) - p_0 = 0 \quad (19)$$

Since exact equality is not achieved in the first trial, a correction is applied to  $p_0$  using the sensitivity information contained in the STM. The corrected initial state is then propagated again, and the process is repeated until the fixed-point condition is satisfied within tolerance.

Finally, in some cases the computation can be simplified by exploiting symmetries. For instance, for orbits around  $L_1$ ,  $L_2$ , and  $L_3$ , symmetries with respect to the  $x$ - $y$  or  $x$ - $z$  planes allow the integration of only half the trajectory. The other half is reconstructed by symmetry. This reduces the number of free variables in the shooting method and accelerates convergence.

### 6.1.1 Newton's Method

Since the difference between the initial and final state is generally non-zero at the first trial, Newton's method is applied to iteratively refine the initial condition [Masdemont and Mondelo, 2004]. Let  $\Phi(t_f, t_0)$  denote the STM propagated along the trajectory from  $t_0$  to  $t_f$ . For a full-period shooting method, the residual function is :

$$F(p_0) = x(t_f) - p_0 \quad (20)$$

with the Jacobian of the system :

$$\frac{\partial F}{\partial p_0} = \Phi(t_f, t_0) - I \quad (21)$$

The Newton update can be written as follow :

$$p_0^{(k+1)} = p_0^{(k)} - [\Phi(t_f, t_0) - I]^{-1} F(p_0^{(k)}) \quad (22)$$

In practice, periodicity is often imposed through a Poincaré map (mentioned in the previous part).

$$P : p_0 \mapsto x(t_{\text{cross}}) \quad (23)$$

The Jacobian of the Poincaré map is obtained by projecting the STM onto the section [Masdemont and Mondelo, 2004]:

$$\frac{\partial P}{\partial p_0} = \left( I - \frac{f(x)n}{nf(x)} \right) \Phi(t_{\text{cross}}, t_0) \quad (24)$$

where  $f(x)$  is the vector field of the CRTBP and  $n$  is the normal to the Poincaré plane. This projection accounts for the fact that only a subset of the state variables (e.g.,  $(x, \dot{x}, z, \dot{z})$ ) is free on the section, while the others are fixed by the crossing condition.

This procedure is repeated until the residual  $\|F(p_0)\|$  falls below a prescribed tolerance. Newton's method exhibits quadratic convergence when the initial guess is sufficiently close to the true solution, which makes it the standard tool for refining periodic orbits in the CRTBP.

### 6.1.2 Special Case of the Initialization around $L_4$

For periodic orbits around the collinear points  $L_1$  and  $L_2$ , initial guesses can be conveniently obtained from publicly available databases maintained by the [Jet Propulsion Laboratory]. In contrast, no such database exists for the triangular points  $L_4$  and  $L_5$ . The initialization around  $L_4$  therefore requires a different approach, based on the eigenstructure of the linearized dynamics at the equilibrium. This approach relies on a modal decomposition, which provides a rigorous way to generate initial guesses for different families of periodic orbits by exciting specific modal directions [Anderlecht, 2016, Greene et al., 2022].

Let  $\xi = (\delta x, \delta y, \delta z, \delta \dot{x}, \delta \dot{y}, \delta \dot{z})^\top$  be the deviation from the equilibrium  $L_4$  in the synodic frame. The CRTBP equations of motion are :

$$\ddot{x} - 2\dot{y} = \Omega_x, \quad \ddot{y} + 2\dot{x} = \Omega_y, \quad \ddot{z} = \Omega_z \quad (25)$$

Linearizing at  $L_4$  gives a first-order system  $\dot{\xi} = A\xi$  with the standard block structure

$$A = \begin{bmatrix} 0_{3 \times 3} & I_3 \\ H & G \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 2 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad H = \begin{bmatrix} \Omega_{xx} & \Omega_{xy} & 0 \\ \Omega_{xy} & \Omega_{yy} & 0 \\ 0 & 0 & \Omega_{zz} \end{bmatrix}_{L_4} \quad (26)$$

Where the Hessian entries of  $\Omega$  are :

$$\Omega_{xx} = 1 - (1 - \mu) \left( \frac{1}{r_1^3} - 3 \frac{(x + \mu)^2}{r_1^5} \right) - \mu \left( \frac{1}{r_2^3} - 3 \frac{(x - 1 + \mu)^2}{r_2^5} \right) \quad (27)$$

$$\Omega_{yy} = 1 - (1 - \mu) \left( \frac{1}{r_1^3} - 3 \frac{y^2}{r_1^5} \right) - \mu \left( \frac{1}{r_2^3} - 3 \frac{y^2}{r_2^5} \right) \quad (28)$$

$$\Omega_{xy} = 3(1 - \mu) \frac{(x + \mu)y}{r_1^5} + 3\mu \frac{(x - 1 + \mu)y}{r_2^5} \quad (29)$$

$$\Omega_{zz} = -\frac{1 - \mu}{r_1^3} - \frac{\mu}{r_2^3} \quad (30)$$

At  $L_4$  we have :

$$(x_{L_4}, y_{L_4}, z_{L_4}) = \left( \frac{1}{2} - \mu, \frac{\sqrt{3}}{2}, 0 \right) \quad (31)$$

$$\Omega_{xx} = \frac{3}{4}, \quad \Omega_{yy} = \frac{9}{4}, \quad \Omega_{xy} = \frac{3\sqrt{3}}{4}(1 - 2\mu), \quad \Omega_{zz} = -1 \quad (32)$$

The  $6 \times 6$  matrix  $A$  has three conjugate pairs of purely imaginary eigenvalues

$$\text{Spec}(A) = \{\pm i\omega_1, \pm i\omega_2, \pm i\omega_3\}, \quad 0 < \omega_1 < \omega_2 < \omega_3 \quad (33)$$

with associated complex eigenvectors  $v_k \in \mathbb{C}^6$ . A real modal basis is obtained by stacking real/imaginary parts:

$$C = [\Re v_1 \ \Im v_1 \ \Re v_2 \ \Im v_2 \ \Re v_3 \ \Im v_3] \in \mathbb{R}^{6 \times 6}, \quad (34)$$

which provides the real change of variables  $\xi = C Z$  and :

$$C^{-1}AC = D = \bigoplus_{k=1}^3 \begin{bmatrix} 0 & -\omega_k \\ \omega_k & 0 \end{bmatrix} \quad (35)$$

For each mode  $k = 1, 2, 3$ , the dynamics in  $(Z_{2k-1}, Z_{2k})$  are a pure rotation:

$$\begin{bmatrix} Z_{2k-1}(t) \\ Z_{2k}(t) \end{bmatrix} = A_k \begin{bmatrix} \cos(\omega_k t + \phi_k) \\ \sin(\omega_k t + \phi_k) \end{bmatrix} \quad (36)$$

Thus,

$$\xi(t) = C Z(t) = \sum_{k=1}^3 A_k \left( \cos(\omega_k t + \phi_k) \Re v_k + \sin(\omega_k t + \phi_k) \Im v_k \right) \quad (37)$$

and map to physical coordinates

$$x_0 = x_{L_4} + C Z_0 \quad (38)$$

Choosing  $k = 1$  (lowest frequency, in-plane) yields a planar Lyapunov-type seed,  $k = 2$  (mainly vertical) yields a vertical/halo-like seed,  $k = 3$  yields a large seed. The phase  $\phi_k$  is used to satisfy a simple constraint (e.g. fix  $x = x_{L_4}$  or impose a section crossing), after which a Newton shooting correction enforces periodicity (or section-to-section closure) in the full nonlinear CRTBP.

| Mode index | Frequency  | Dominant direction    | Orbit family             |
|------------|------------|-----------------------|--------------------------|
| 0          | $\omega_1$ | Planar short $(x, y)$ | Small Lyapunov           |
| 1          | $\omega_2$ | Vertical $(z)$        | Vertical Lyapunov / Halo |
| 2          | $\omega_3$ | Planar long $(x, y)$  | Large Lyapunov           |

Table 1: Classification of modes and orbit families around  $L_4$

### 6.1.3 Computation of Families of Periodic Orbits

Once a single periodic solution has been computed, continuation techniques are employed to generate entire families of orbits [Chupin, 2016]. The principle is to vary a selected parameter in small steps in order to remain within the convergence range of Newton's method. Then, the previously corrected solution is used as the initial guess for the next



step, and the Newton shooting method is applied until the periodicity condition is restored (figure 12).

Several variables can serve as continuation parameters, such as the initial displacement in the  $x$ ,  $y$ ,  $z$  directions or the Jacobi constant. The choice depends on the family of interest and numerical constraints. In this project, we chose to vary  $x$  around  $L_1$  and  $y$  around  $L_4$ .

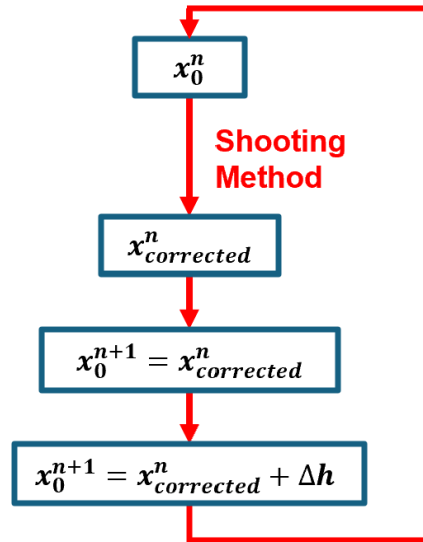
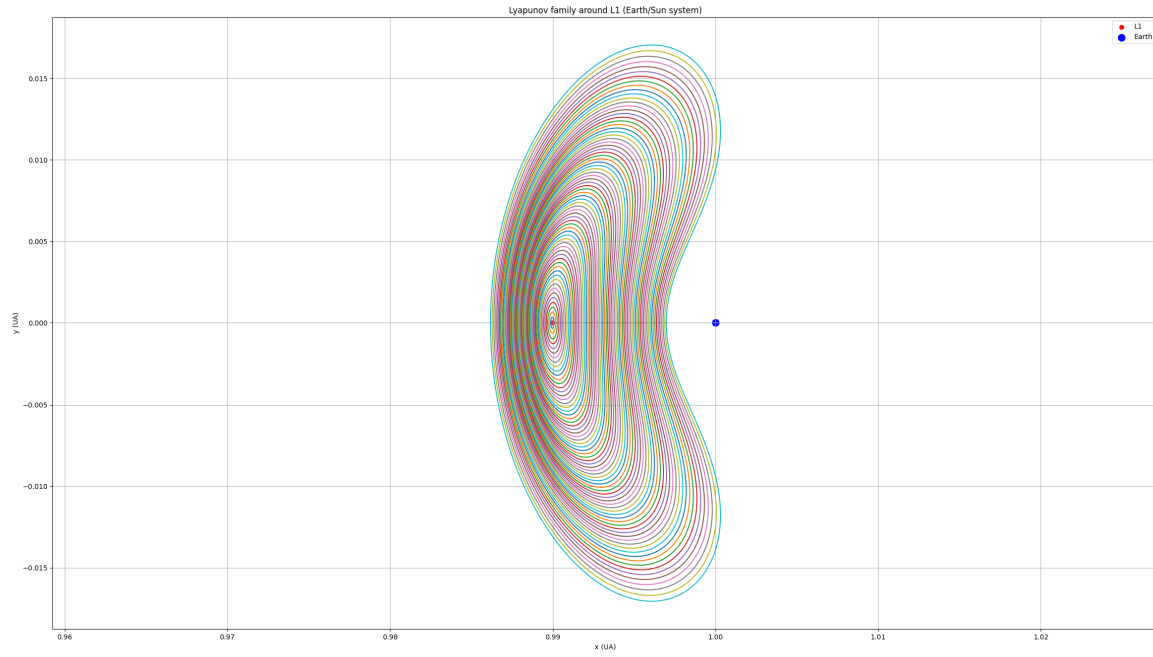
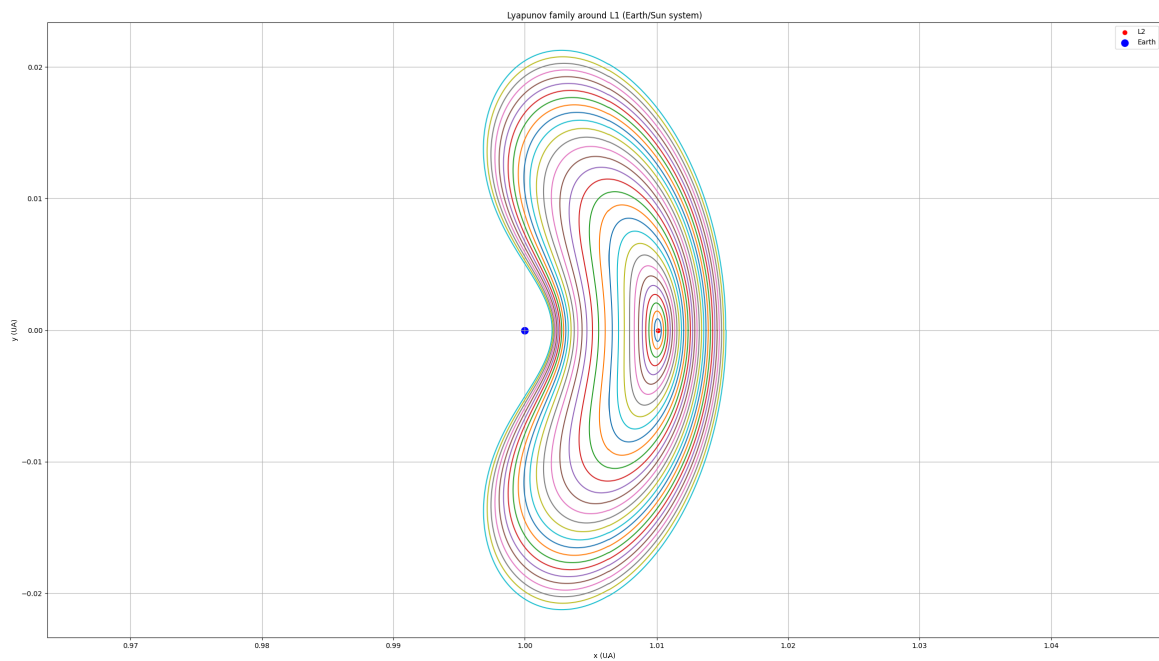
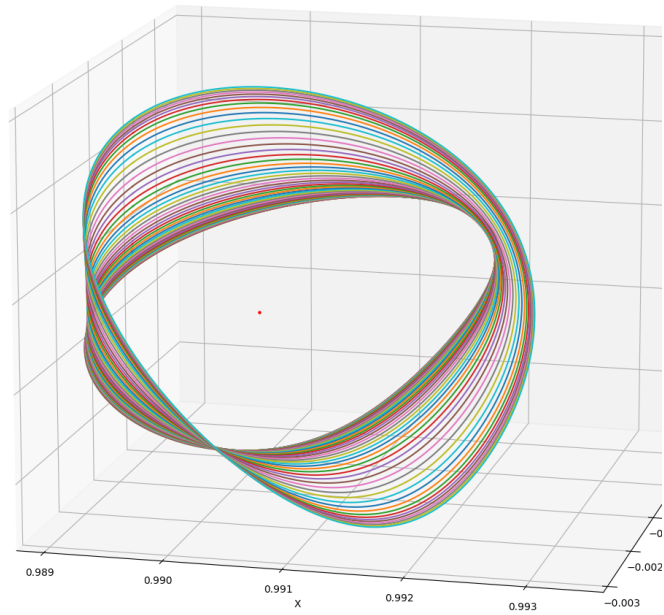
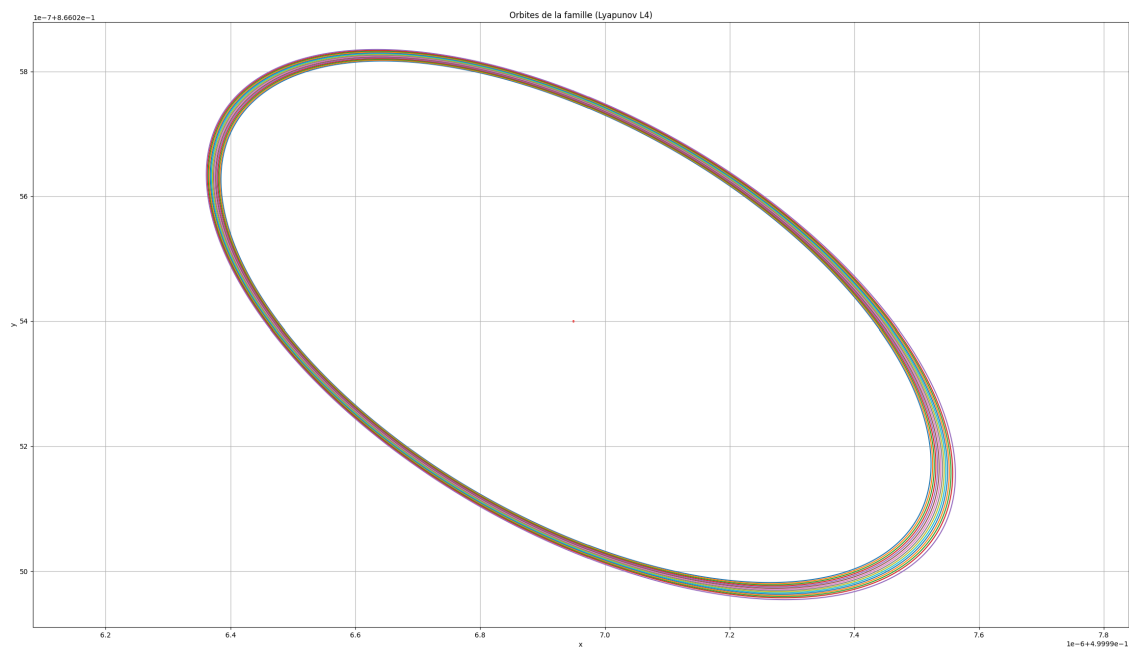


Figure 12: Continuation procedure around  $L_1$

This process produces a continuous sequence of periodic orbits that form a family. In many cases, continuation reveals bifurcations. The continuation method provides a systematic way to explore the solution space without requiring an independent guess for each orbit. These families can be observed in Figures 14–16.

Figure 13: Planar Lyapunov Family around  $L_1$ Figure 14: Planar Lyapunov Family around  $L_2$

Figure 15: Halo Family around  $L_1$ Figure 16: Planar Lyapunov Family around  $L_4$

## 7 Trajectory Design

Now that periodic orbits around  $L_1$  and  $L_4$  have been computed, the next step is the design of transfer trajectories linking these libration regions. The idea is to exploit the dynamical pathways naturally provided by the invariant manifolds of periodic orbits. By departing from a Lyapunov orbit at  $L_1$  and following its unstable manifold, the spacecraft can escape the vicinity of  $L_1$  with minimal effort and evolve along the Sun/Earth system. The arrival near  $L_4$ , however, requires a different approach due to its intrinsic stability.

### 7.1 Transfer Strategy

The transfer between two libration point orbits cannot be achieved through simple two-body maneuvers. Instead, it must exploit the geometric structure of the CRTBP. Each periodic orbit around an unstable equilibrium point, such as  $L_1$ , is associated with a family of stable and unstable invariant manifolds that govern the motion in its neighborhood [Koon et al., 2022].

The unstable manifold gathers trajectories that naturally drift away from the orbit, while the stable manifold gathers those that converge toward it. These manifolds act as the boundaries between dynamical regions, forming “tubes” that connect neighborhoods without continuous thrusting. By injecting a spacecraft into the unstable manifold of a Lyapunov orbit around  $L_1$  and the right direction, the natural dynamics drive the trajectory away from the Earth and along the Sun/Earth line, as shown in figure 17.

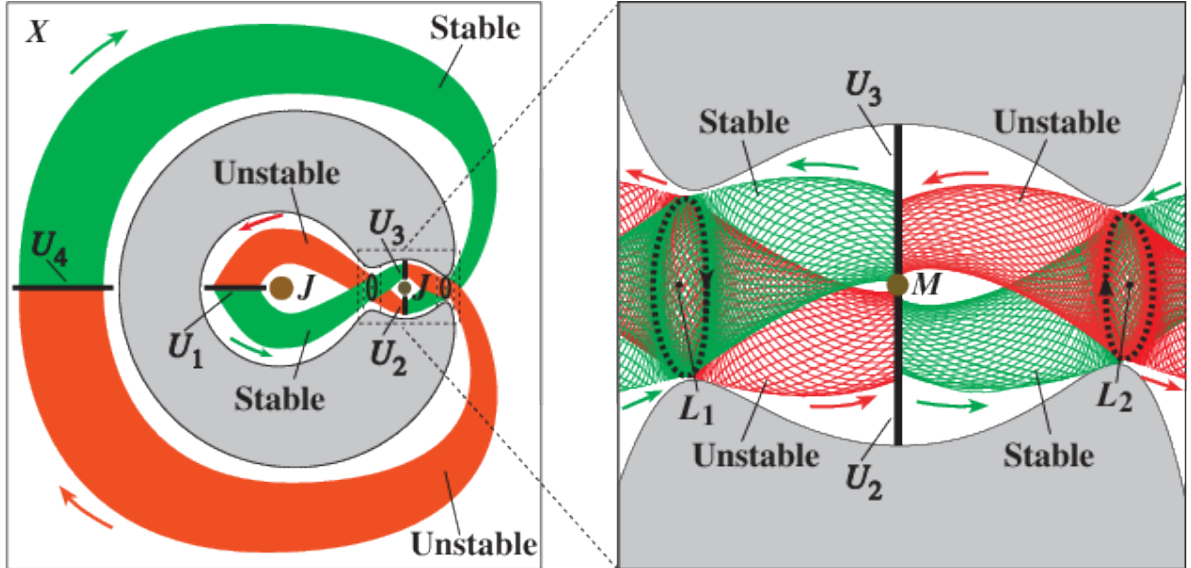


Figure 17: Invariant Manifolds in the CRTBP [Gomez et al., 2001], Jupyter/moon system (the pattern is the same in Earth/Sun system)

From a practical perspective, the unstable manifold of  $L_1$  is computed numerically using the unstable direction from the variational equations and the equations of motion. Perturbations along the unstable eigenvector of the monodromy matrix provide the initial displacement for manifold propagation. Once inside, the resulting trajectories form structures that guide the motion toward the outer regions of the system.

Unlike  $L_1$ , the triangular point  $L_4$  is linearly stable in the Sun/Earth system and does not possess classical invariant manifolds. The transfer relies on the natural dynamics of the CRTBP to approach the vicinity of  $L_4$ , followed by a small impulsive maneuver to match the spacecraft's energy (Jacobi constant) with the one of the Lyapunov orbit around  $L_4$ .

The transfer trajectory follows a four-step procedure. The spacecraft departs from a planar Lyapunov orbit around  $L_1$  by a small perturbation along its unstable eigenvector, placing it onto the corresponding unstable manifold. The trajectory is then propagated until it intersects a predefined cutting line located about  $30^\circ$  ahead of the Sun/Earth line. At this point, a first impulsive maneuver  $\Delta V_1$  is applied to adjust the velocity and energy so that the trajectory naturally evolves toward the vicinity of  $L_4$  under the dynamics of the CRTBP. Upon arrival near  $L_4$ , a second impulsive maneuver  $\Delta V_2$  is performed to match the Lyapunov orbit.

## 7.2 Matching Method Using Newton Iteration

To compute the transfer trajectory linking the Lyapunov orbits around  $L_1$  and  $L_4$ , a shooting-like Newton method is employed. The approach consists on propagating two trajectories simultaneously: one departing forward in time from the Lyapunov orbit around  $L_1$  along its unstable direction, and another propagated backward in time from the Lyapunov orbit around  $L_4$  (after a  $\Delta V_2$ ). The goal is to adjust both trajectories so that they intersect at the same point on the predefined cutting line (shown in figure 18).

Each trajectory is initialized by a small perturbation applied along the appropriate direction. For  $L_1$ , a positive perturbation  $\varepsilon_1$  is applied along the unstable eigenvector of the orbit's monodromy matrix. For  $L_4$ , a negative perturbation  $\varepsilon_2$  is applied along the velocity direction on the Lyapunov orbit. Both trajectories are then integrated until they cross the cutting line, where their states are compared.

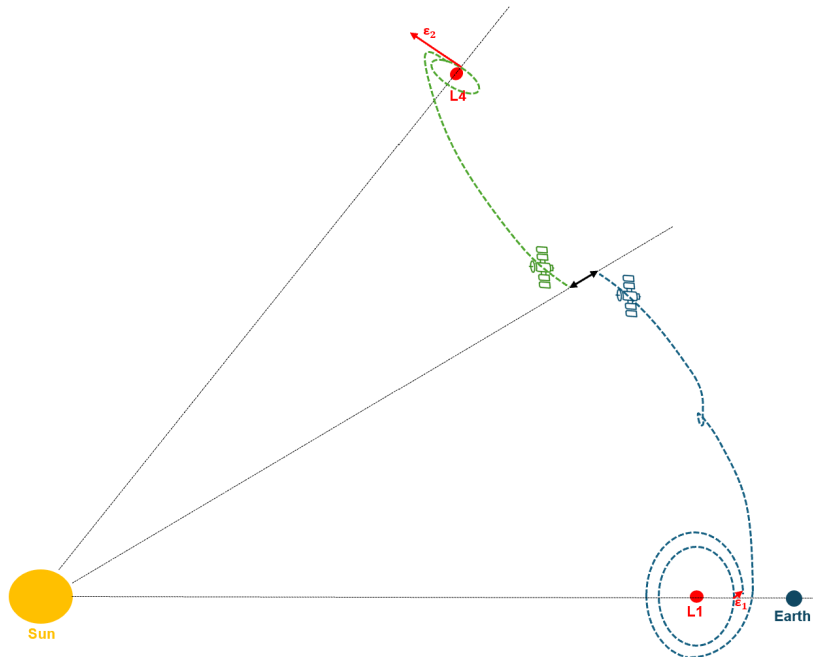


Figure 18: Matching method

Let  $x_1(\varepsilon_1)$  and  $x_2(\varepsilon_2)$  denote the intersection states of the forward and backward propagation. The objective is to reduce to zero the position mismatch on the cutting line:

$$F(\varepsilon_1, \varepsilon_2) = x_1(\varepsilon_1) - x_2(\varepsilon_2) = 0 \quad (39)$$

To correct the initial perturbations, the state transition matrices  $\Phi_1$  and  $\Phi_2$  propagated along both trajectories are used to linearize the mapping between the initial displacements and the intersection states. At each iteration, the correction vector is obtained by solving:

$$\begin{bmatrix} \Phi_1 & -\Phi_2 \end{bmatrix} \begin{bmatrix} \delta\varepsilon_1 \\ \delta\varepsilon_2 \end{bmatrix} = -F(\varepsilon_1, \varepsilon_2) \quad (40)$$

The updated perturbations are :

$$\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \end{bmatrix}_{k+1} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \end{bmatrix}_k + \begin{bmatrix} \delta\varepsilon_1 \\ \delta\varepsilon_2 \end{bmatrix} \quad (41)$$

and the process is repeated until the distance between the two trajectories on the cutting line falls below a prescribed tolerance.

### 7.3 Python Results

The theoretical formulation of the matching procedure is now implemented in Python to verify its effectiveness in generating feasible transfer trajectories. This part provides a first-order validation of the transfer geometry before introducing a higher-fidelity force model in GMAT. We work in the nondimensional CRTBP scaling with Sun/Earth primaries:

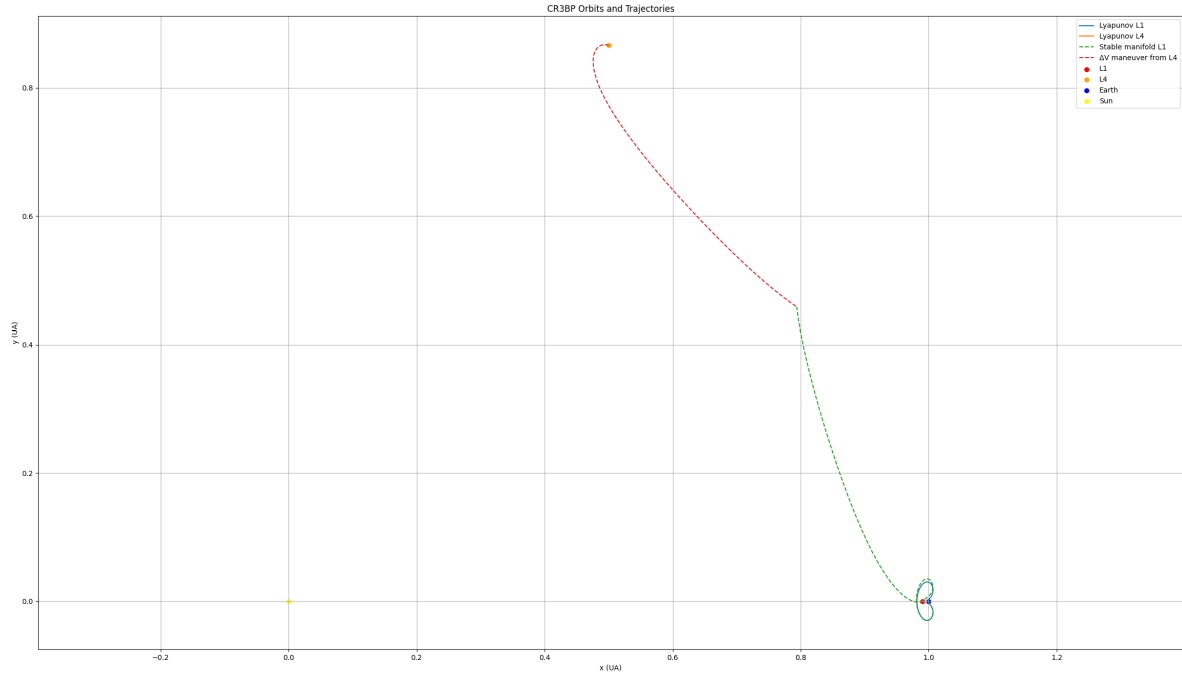
| Earth/Sun System         |                                            |
|--------------------------|--------------------------------------------|
| Parameter                | Value                                      |
| Mass parameter ( $\mu$ ) | $3.0542 \times 10^{-6}$                    |
| Time unit (TU)           | 5 022 635 s                                |
| Length unit (LU)         | 149 597 871 km                             |
| L1 position (LU)         | [0.98997092, 0, 0]                         |
| L4 position (LU)         | [0.49999695, 0.86602540, 0]                |
| Integrator               | Runge–Kutta (RK5(4,5))                     |
| Relative tolerance       | $1 \times 10^{-12}$                        |
| Absolute tolerance       | $1 \times 10^{-12}$                        |
| Cutting line             | $y = \tan \theta_c x, \theta_c = 30^\circ$ |

Table 2: Constants used in the Sun/Earth system

| <b>Lyapunov L1</b>                       |                                                                                   |
|------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Parameter</b>                         | <b>Value</b>                                                                      |
| Position 0 (LU)                          | $[1.010346, 4.185673 \times 10^{-22}, -5.736385 \times 10^{-26}]$                 |
| Velocity 0 (LU/TU)                       | $[9.066985 \times 10^{-16}, -1.711167 \times 10^{-3}, -2.137685 \times 10^{-25}]$ |
| Period(TU)                               | 3.0560974087923287                                                                |
| Poincaré section                         | $y, z = 0$ and $x \neq 0$ with $\dot{y} > 0$ and $\dot{x} = 0$                    |
| Tolerance                                | $1 \times 10^{-12}$                                                               |
| <b>Halo L1</b>                           |                                                                                   |
| Position 0 (LU)                          | $[0.988816, -1.447777 \times 10^{-25}, 4.035716 \times 10^{-4}]$                  |
| Velocity 0 (LU/TU)                       | $[2.680859 \times 10^{-16}, 8.873463 \times 10^{-3}, -1.036203 \times 10^{-16}]$  |
| Period(TU)                               | 3.0600044536535105                                                                |
| Poincaré section                         | $y = 0$ and $x, z \neq 0$ with $\dot{y} > 0$ and $\dot{x} = 0$                    |
| Tolerance                                | $1 \times 10^{-12}$                                                               |
| <b>Lyapunov L4</b>                       |                                                                                   |
| Amplitude/Mode/Phase ( $\phi$ )          | $1 \times 10^{-3}$ / Small / $0^\circ$                                            |
| Position 0 (LU)                          | $[0.499997, 0.866376, 1.90303 \times 10^{-21}]$                                   |
| Velocity 0 (LU/TU)                       | $[0.00057008, -0.000227807, 1.17045 \times 10^{-20}]$                             |
| Period(TU)                               | 6.283250327055708                                                                 |
| Poincaré section                         | $x = X_{L4}, z = 0$ and $y \neq y_{L4}$ with $\dot{y} > 0$                        |
| Tolerance                                | $1 \times 10^{-10}$                                                               |
| <b>Matching method</b>                   |                                                                                   |
| Initial perturbation L1 ( $\epsilon_1$ ) | 0.001                                                                             |
| Initial $\Delta V$ L4 ( $\epsilon_2$ )   | 0.01                                                                              |
| Tolerance                                | $1 \times 10^{-9}$                                                                |

Table 3: Constants used for trajectory computation

Using the constants summarized in Table 2 and Table 3, the numerical propagation of the two Lyapunov orbits and the subsequent matching procedure yield the following transfer trajectory.

Figure 19: Reference transfer trajectory from  $L_1$  to  $L_4$  (CRTBP model)

| Parameter                        | Value       |
|----------------------------------|-------------|
| Perturbation L1 ( $\epsilon_1$ ) | 0.001481963 |
| $\Delta V$ L4 ( $\epsilon_2$ )   | 0.034064286 |
| Total $\Delta V$ (km/s)          | 4.73889     |

Table 4: Python results

Under the impulsive-thrust assumption, the propellant mass required for the transfer can be estimated with the Tsiolkovsky rocket equation. Let  $m_0$  be the initial (wet) mass,  $m_f$  the final (dry + residuals) mass after the maneuver,  $I_{sp}$  the engine specific impulse, and  $g_0 = 9.80665 \text{ m s}^{-2}$  the standard gravity. The effective exhaust velocity is  $v_e = I_{sp}g_0$ . For two sequential burns  $\Delta V_1$  and  $\Delta V_2$  (Table 4), the exact mass is :

$$m_1 = m_0 \exp\left(-\frac{\Delta V_1}{v_e}\right), \quad (42)$$

$$m_f = m_1 \exp\left(-\frac{\Delta V_2}{v_e}\right) = m_0 \exp\left(-\frac{\Delta V_1 + \Delta V_2}{v_e}\right), \quad (43)$$

so the total propellant is :

$$m_{\text{prop}} = m_0 - m_f = m_0 \left[1 - \exp\left(-\frac{\Delta V_1 + \Delta V_2}{I_{sp}g_0}\right)\right]. \quad (44)$$



| Parameter               | Value     |
|-------------------------|-----------|
| Initial mass ( $m_0$ )  | 2500 kg   |
| Dry mass                | 1160 kg   |
| Specific impulse (Isp)  | 300s      |
| Total $\Delta V$ (km/s) | 4.73889   |
| Propellant mass         | 2000.7 kg |

Table 5: Propellant mass

The estimated propellant mass required for the transfer is significantly higher (Table 5) than the initial allocation for the EOS-L4 mission (1300 kg). This discrepancy suggests that, although the current approach exploits the dynamical structure of the CRTBP, it remains suboptimal in terms of energy efficiency. Further optimization is therefore necessary to reduce the total  $\Delta V$  cost and bring the transfer within the mission's propellant constraints.

To further investigate the behavior of the transfer, a continuation method was applied on the initial perturbation parameter  $\varepsilon_1$  defining the departure along the unstable manifold of the  $L_1$  Lyapunov orbit. In this process,  $\varepsilon_1$  was gradually varied from 0.8 its nominal value to 1.2 that value. For each new value of  $\varepsilon_1$ , the transfer trajectory was recomputed using the Newton matching procedure described previously. This approach allows assessing the sensitivity of the transfer geometry with respect to the initial displacement from  $L_1$ , and provides a continuous family of solutions illustrating how the unstable manifold deforms and connects to the  $L_4$  region under small perturbations.

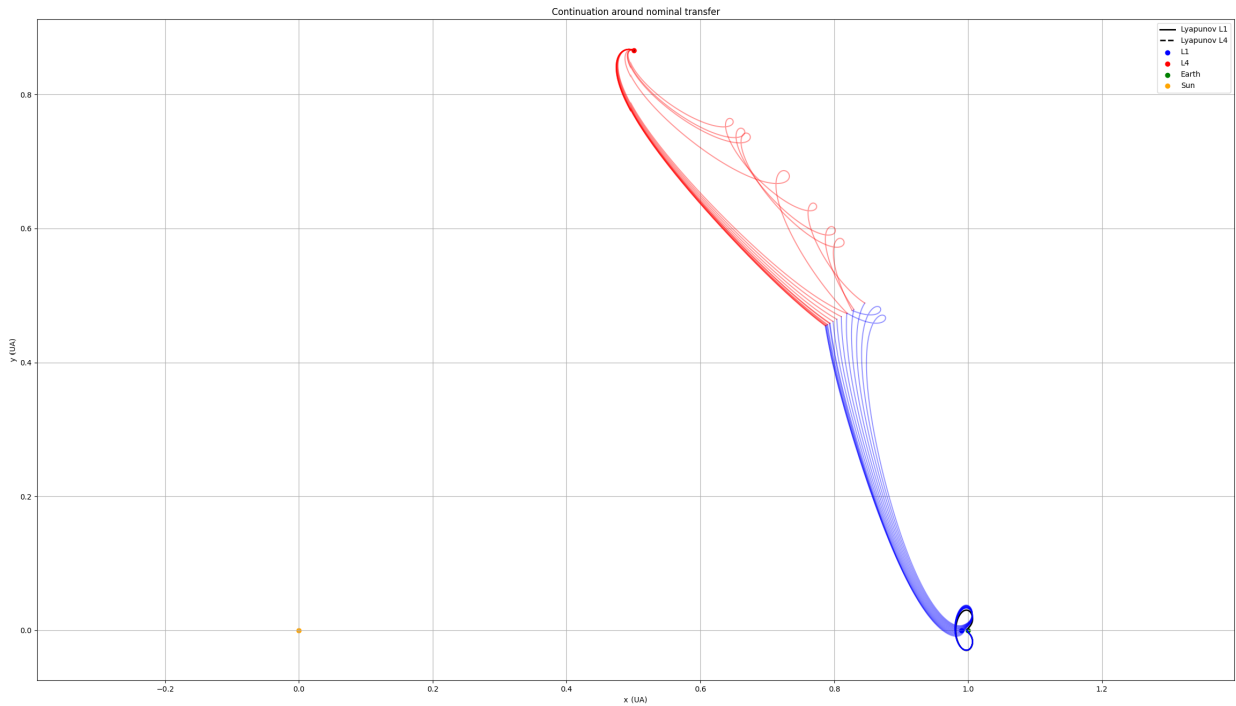
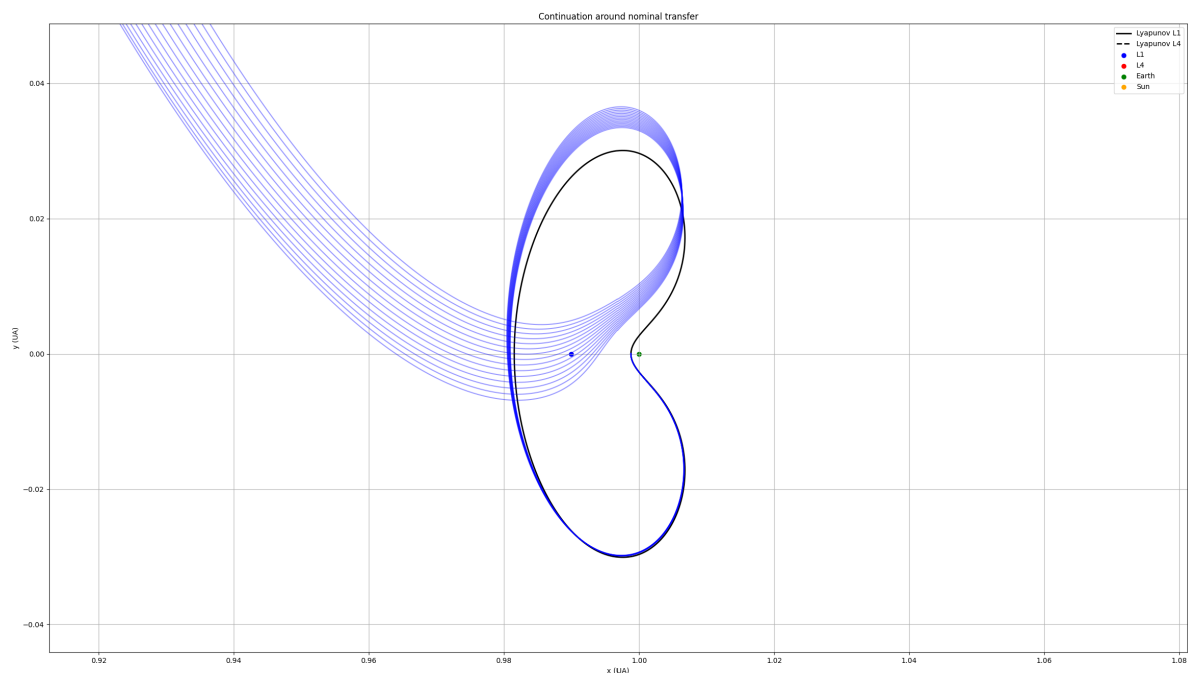


Figure 20: Trajectory with continuity method (eps1)

On the  $L_1$  side, the continuation process produces the following pattern. Each curve corresponds to a different value of the perturbation parameter  $\varepsilon_1$ .

Figure 21: Trajectory pattern near  $L_1$ 

To quantify the impact of the  $L_1$  injection on the maneuvering cost, the total  $\Delta V$  was evaluated for each member of the family generated by the continuation on  $\varepsilon_1$ . The results show a clear trend: as the initial perturbation  $\varepsilon_1$  increases, the overall  $\Delta V$  required for the transfer decreases. A larger displacement along the unstable manifold effectively injects the spacecraft onto a trajectory that departs more efficiently from the  $L_1$  region, thereby reducing the energy needed to reach the vicinity of  $L_4$ . It should be noted, however, that this trend does not extend indefinitely. Beyond a certain threshold, the continuation process diverges and the periodicity or matching conditions can no longer be satisfied, indicating the breakdown of the linear approximation used for initializing the manifold. In addition, the present analysis neglects the small impulsive  $\Delta V$  that would be required to precisely enter the unstable manifold, meaning that the reported values represent an idealized lower bound for the total transfer cost.

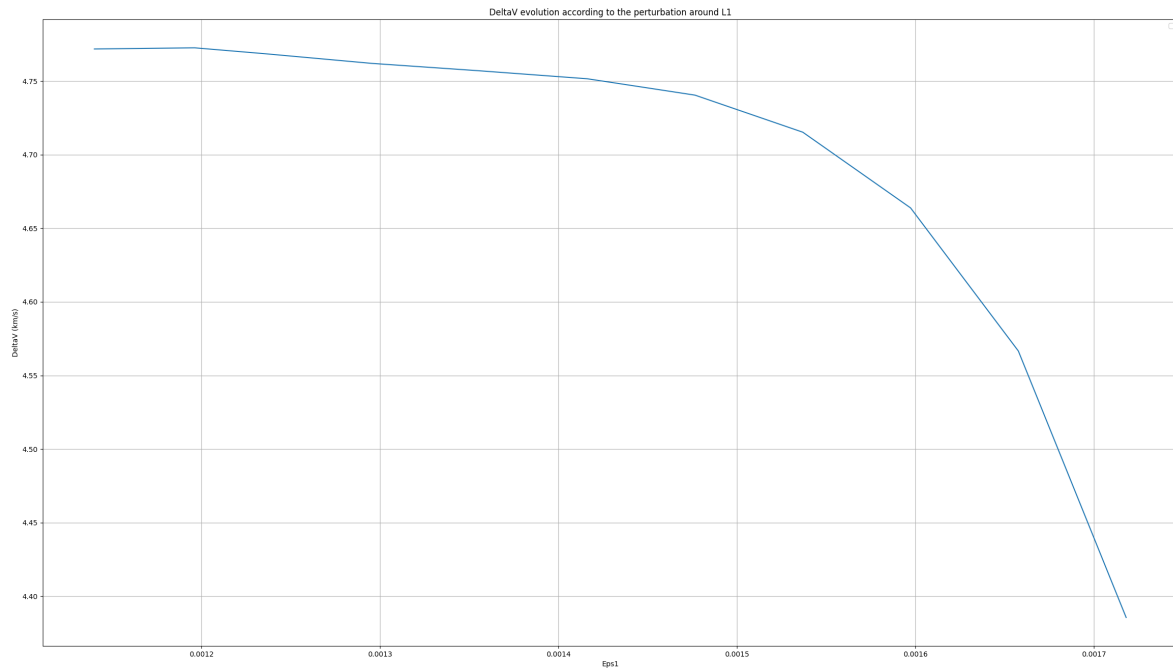


Figure 22: Total  $\Delta V$  as a function of the manifold injection parameter  $\varepsilon_1$

We next assessed the influence of the cutting-line angle  $\theta_c$  (measured from the Sun/Earth axis) on the  $\Delta V$ .

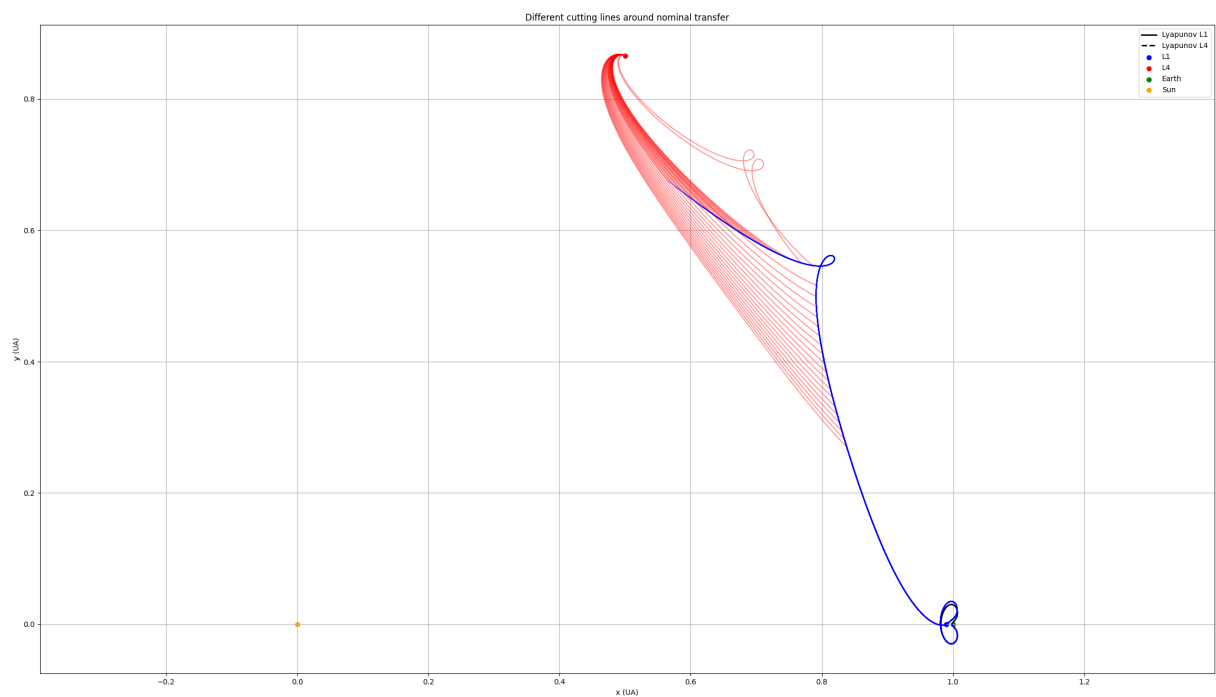


Figure 23: Trajectory pattern in function of the cutting line inclination

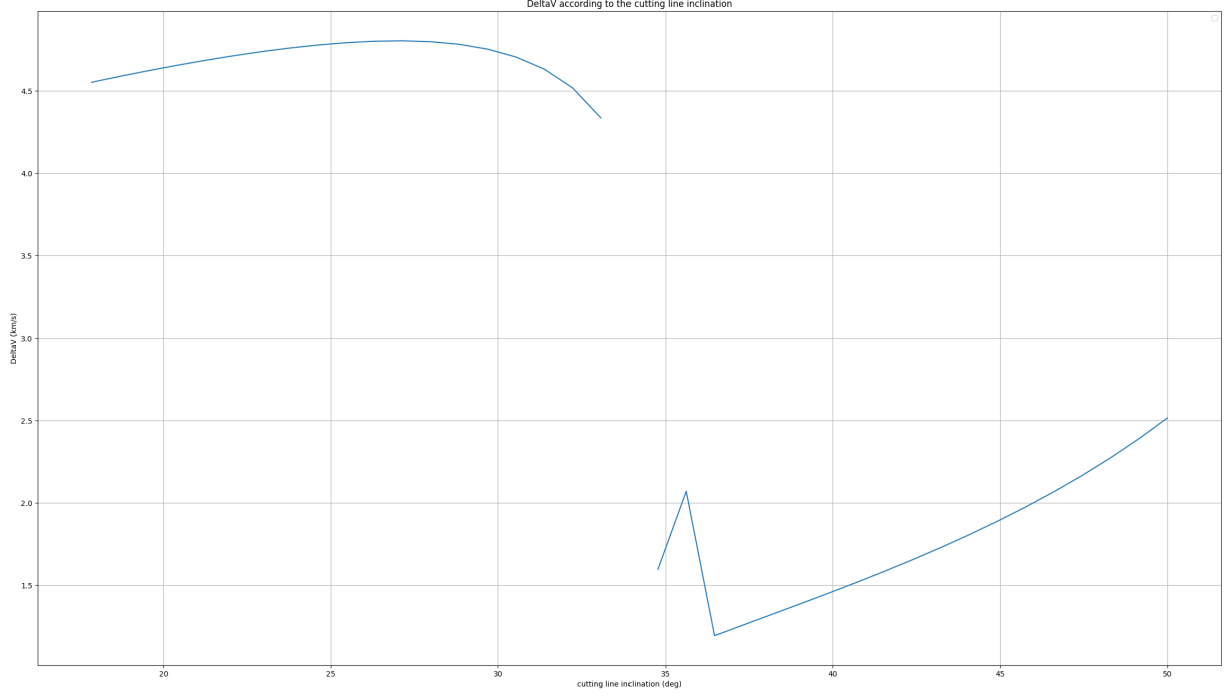
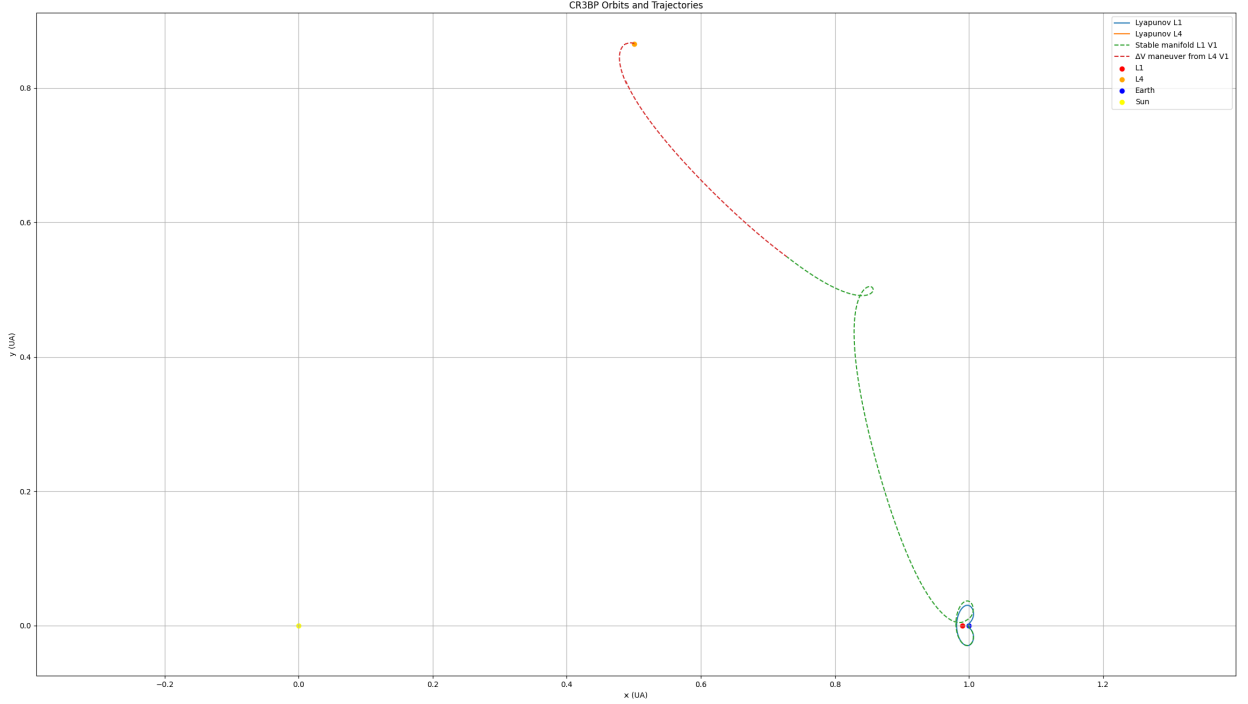


Figure 24: Total  $\Delta V$  as a function of the cutting line inclination

The geometry of the transfer trajectory changes significantly with the inclination of the cutting line  $\theta_c$ . Unlike the perturbation parameter  $\varepsilon_1$ , whose effect on the total  $\Delta V$  is relatively smooth, the dependence on  $\theta_c$  is highly nonlinear. As the cutting line rotates, the intersection point with the unstable manifold of  $L_1$  shifts considerably, leading to abrupt variations in the required  $\Delta V$ . Among the tested configurations, an inclination of approximately  $37^\circ$  was found to minimize the total maneuvering cost. This result is physically consistent: at this angle, the cutting line becomes nearly tangent to the outgoing trajectory from the  $L_1$  manifold, reducing the velocity mismatch at the intersection and thereby minimizing the corrective impulse required for the transfer toward  $L_4$ .

Finally, combining the continuation on  $\varepsilon_1$  and the selection of a suitable cutting-line inclination  $\theta_c$ , an optimized transfer was identified. We obtain the resulting trajectory and results.

Figure 25: Optimized  $L_1 \rightarrow L_4$  transfer in the synodic frame.

| Parameter                        | Value       |
|----------------------------------|-------------|
| Initial mass ( $m_0$ )           | 2500 kg     |
| Dry mass                         | 1160 kg     |
| Specific impulse (Isp)           | 300s        |
| Cutting line inclination (deg)   | $37^\circ$  |
| Perturbation L1 ( $\epsilon_1$ ) | 0.001778356 |
| $\Delta V$ L4 ( $\epsilon_2$ )   | 0.028883262 |
| Total $\Delta V$ (km/s)          | 1.008056    |
| Propellant mass                  | 725.3 kg    |

Table 6: Result after optimization

The transfer design yields a preliminary propellant mass of approximately 725 kg (Table 6), which is lower than the mission baseline of 1300 kg. This result is encouraging, but must be interpreted with caution: the estimate is optimistic because several contributions have been neglected in the present analysis. In particular, we have not accounted for the  $\Delta V$  required to inject the spacecraft into the unstable manifold, the insertion burn at the departure Lyapunov orbit around  $L_1$ , station-keeping for the operational lifetime at  $L_4$ , nor margins for guidance, navigation and control or finite-burn losses. We also assumed the use of a high-performance chemical engine in vacuum (approximately  $I_{sp} = 300$  s), which should be regarded as optimistic. In practice, monopropellant thrusters (hydrazine) typically deliver lower specific impulse (on the order of 220–230 s). Moreover, propellants such as hydrazine are toxic and corrosive, introducing integration, safety, and cost constraints. Alternative options—green propellants, different bipropellant architectures, or electric propulsion with much higher  $I_{sp}$ —should be considered during final trade studies to improve performance.

To confirm and refine this preliminary result, the next step consists of validating the transfer in a high-fidelity environment, the matched trajectory will be propagated and optimized in GMAT.

## 7.4 GMAT Results

While the Python simulations provide valuable insight into the underlying dynamics of the CRTBP, they remain limited by the assumptions of the idealized three-body model. To assess the robustness of the designed transfer under more realistic conditions, the same trajectory is reproduced in the General Mission Analysis Tool (GMAT).

GMAT is an open-source astrodynamics software developed by the GSFC [Hughes, 2007]. It provides a high-fidelity environment for spacecraft trajectory design, orbit propagation, and mission analysis. Unlike analytical or semi-analytical models such as those implemented in Python for the CRTBP, GMAT integrates the full equations of motion in an inertial frame and supports advanced numerical schemes, precise force modeling (including perturbations, solar radiation pressure and multiple-body effects).

The GMAT simulation was based on the same initial conditions and constants listed in Table 2. Several modifications were introduced to replicate the Newton matching procedure and the propagation scheme described previously [Hughes, 2007]:

- Two spacecraft are created: one departing from the Lyapunov orbit around  $L_1$ , and one initialized near  $L_4$ . This configuration allows forward and backward propagation to be performed simultaneously within the same simulation.
- The built-in *Propagate()* function in GMAT is used to advance each spacecraft until a predefined geometric condition is met, such as the intersection with the cutting line.
- A high-order Runge–Kutta integrator (RungeKutta89) is selected to ensure high numerical accuracy, with relative and absolute tolerances set to  $10^{-1}m$ .
- The gravitational influence of the Moon is included in the simulation, providing a more realistic dynamical environment than in the idealized CRTBP.
- Epoch is fixed at 5 december 2005
- The transfer is modeled using two instantaneous  $\Delta V$  maneuvers, corresponding respectively to the injection from the  $L_1$  manifold and the final insertion near the  $L_4$  periodic orbit.

Unlike in the simplified CRTBP framework, the Sun/Earth dynamics in GMAT include the true orbital eccentricity of the primaries, third-body perturbations, and epoch-dependent initial conditions. As a result, the relative configuration of the Sun, Earth, and libration regions ( $L_1$ ,  $L_2$ , and  $L_4$ ) evolves periodically rather than remaining fixed in a perfectly rotating frame. This introduces an “accordion” effect in the geometry of the transfer: the distance and alignment between the libration points oscillate slightly over time. Furthermore, the gravitational influence of the Moon contributes an additional

perturbation that modifies the timing and spatial location of the intersection with the cutting line.

Consequently, the trajectory obtained in GMAT is not expected to match exactly the one generated under the CRTBP assumptions (shown in figure 26). The inclusion of the true orbital eccentricity of the primaries, the lunar perturbation, and the epoch-dependent configuration of the system leads to a noticeable geometric deviation. The simulation gives a  $\Delta V$  of **2.90 km/s**. Nevertheless, the overall dynamical behavior (departure from the  $L_1$  manifold and convergence toward the  $L_4$  region) remains consistent, confirming the qualitative robustness of the transfer concept even within a higher-fidelity dynamical environment. The resulting GMAT trajectory is shown below.

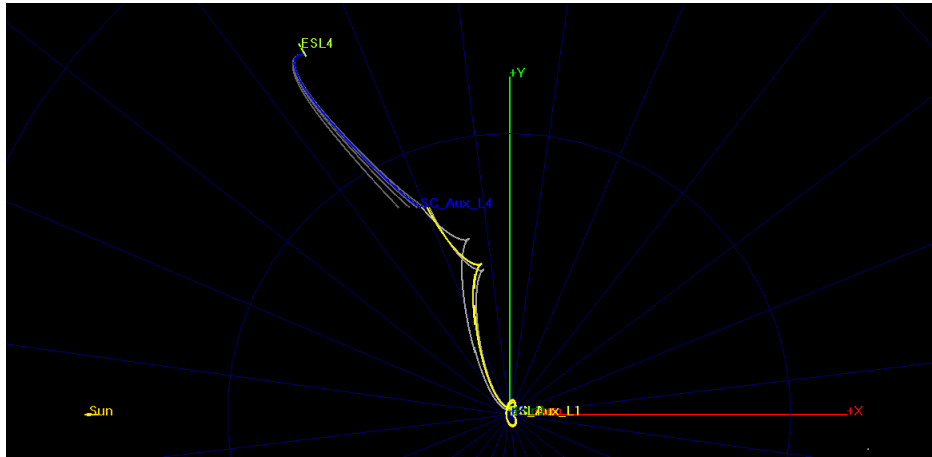


Figure 26: EOS-L4 Trajectory on GMAT

The comparison between the simplified CRTBP model and the high-fidelity GMAT simulations confirms the general consistency of the transfer geometry and  $\Delta V$  budget. The results also reveal the influence of higher-order perturbations and emphasize the need for refined optimization in a future mission scenario. These findings are summarized in the following part.

## 8 Conclusion

This report presented the design, computation, and validation of transfer trajectories between libration point orbits in the Sun/Earth system, focusing on the connection between the unstable manifold of a Lyapunov orbit around  $L_1$  and the stable region near  $L_4$ .

The methodology combined two complementary approaches. The first, based on the CRTBP, provided analytical insight into the dynamical structure of the system and enabled the construction of manifold-assisted transfers through a Newton-type matching method. The second, performed in GMAT, reproduced the same trajectory within a high-fidelity model including the true ephemerides of the Sun, Earth, and Moon, as well as the non-circularity of their orbits.

The comparison between both models revealed that while the global dynamical behavior remains consistent, significant geometric deviations appear once higher-order perturbations are introduced. The inclusion of orbital eccentricity produces an accordion effect, periodically altering the relative distances between the Sun, Earth, and the libration points. The Moon's gravitational perturbation and the epoch-dependent orientation of the system further modify the intersection geometry, leading to a huge  $\Delta V$  difference in the transfer characteristics.

The possible next phase of this project will focus on refining and validating the transfer design in a high-fidelity environment and on introducing systematic optimization into the workflow. In practice this means several concrete actions:

- Implement an automated optimization loop in GMAT to vary launch conditions, manifold injection parameters and cutting-line geometry, and to minimize a cost function.
- Explore multiple Lyapunov seeds around  $L_1$  and larger-amplitude Lyapunov orbits near  $L_4$  to identify low-cost configurations and to build a small database of candidate initial states.
- Extend the study to include transfers from Earth (or a near-Earth parking orbit) to the  $L_1$  Lyapunov manifold and from Earth directly to the  $L_1$  departure phase.
- Test other periodic and quasi-periodic solutions (halo, vertical Lyapunov, quasi-periodic tori) as departure or arrival options to evaluate trade-offs in accessibility and station-keeping.
- Incorporate ephemerides, lunar perturbations, solar radiation pressure, finite-burn modeling, navigation errors and operational margins to obtain a realistic propellant estimate and to design station-keeping strategies for the operational phase at  $L_4$ .
- Perform propulsion trades (chemical architectures, green propellants, electric propulsion) and assess guidance/control strategies (impulsive vs low-thrust transfers, multiple-burn schemes) to improve performance.

These activities will close the loop between theoretical CRTBP-based design and operational mission analysis, producing a validated, optimized transfer solution and a revised, defensible propellant budget for EOS-L4.



## Glossary

**CONOPS** Concept of Operations. 14

**CRTBP** Circular Restricted Three-Body Problem. 3, 4, 9, 10, 12, 14, 16, 18, 23–25, 27, 28, 33–35, 39, 40, 42

**EOS-L4** Mission concept around L4. 14, 28, 34, 35, 39, 40

**GMAT** General Mission Analysis Tool. 1, 3, 7, 9, 25, 33–35, 39

**GSFC** Goddard Space Flight Center. 2, 5–9, 33, 39

**JWST** James Webb Space Telescope. 4

**LEOP** Launch and Early Orbit Phase. 14

**NASA** National Aeronautics and Space Administration. 2–9

**SOHO** Solar and Heliospheric Observatory. 4

**STM** State Transition Matrix. 17

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## 9 Appendix

The following section presents a complementary study, carried out in parallel to the main project, with the aim of developing a deeper understanding of the local dynamics near libration points in the CRTBP. This work extends the classical model by including the acceleration contribution of a solar sail and follows the formulation presented by [Farrés et al., 2019] and related works on sail-stabilized orbits in the Sun–Earth system.

Although this appendix is not directly related to the EOS-L4 mission, it was performed as a self-training exercise to reproduce and interpret published results. The objective is to analyze the linearized dynamics around equilibrium points under the combined influence of gravitational and solar-radiation forces.

The formulation below summarizes the governing equations, the linearization procedure, and the construction of stable and unstable directions used for this analysis. We consider the CRTBP with the Sun and Earth as primaries, and a solar sail spacecraft as the third, massless body. In the synodic rotating frame, the equations of motion are:

$$\ddot{x} - 2\dot{y} = \frac{\partial \Omega}{\partial x} + a_x \quad (45)$$

$$\ddot{y} + 2\dot{x} = \frac{\partial \Omega}{\partial y} + a_y \quad (46)$$

$$\ddot{z} = \frac{\partial \Omega}{\partial z} + a_z \quad (47)$$

where  $\Omega(x, y, z)$  is the effective potential:

$$\Omega(x, y, z) = \frac{1}{2}(x^2 + y^2) + \frac{1 - \mu}{r_{ps}} + \frac{\mu}{r_{pe}} \quad (48)$$

with:

$$r_{ps} = \sqrt{(x + \mu)^2 + y^2 + z^2}, \quad r_{pe} = \sqrt{(x + \mu - 1)^2 + y^2 + z^2} \quad (49)$$

The acceleration due to the solar sail ( $a_s = (a_x, a_y, a_z)$ ) is given by:

$$\vec{a}_s = \beta(1 - \mu) \frac{\cos^2(\alpha)}{r_{ps}^2} [\vec{n} \cos(\alpha) + \vec{r}_s(1 - \cos(\alpha))] \quad (50)$$

where:

- $\beta$  is the sail lightness number (acceleration/gravitational ratio),
- $\alpha$  is the pitch angle (angle between the normal and the sunlight direction),
- $\delta$  is the clock angle (rotation around the sunlight axis),
- $\vec{r}_s$  is the unit vector from the Sun to the sail.

The normal vector  $\vec{n}$  is decomposed in a real basis:

$$\vec{n} = \cos(\alpha)\vec{r}_s + \sin(\alpha) (\cos(\delta)\vec{q} + \sin(\delta)\vec{p}) \quad (51)$$

where  $\vec{q}$  and  $\vec{p}$  are perpendicular vectors constructed via:

$$\vec{q} = \frac{\vec{r}_s \times \hat{z}}{\|\vec{r}_s \times \hat{z}\|}, \quad \vec{p} = \vec{r}_s \times \vec{q} \quad (52)$$

To linearize the system near an equilibrium point such as SL1, we write the state vector as:

$$\vec{X} = (x, y, z, \dot{x}, \dot{y}, \dot{z})^T \quad (53)$$

and compute the Jacobian matrix  $A = Df(\vec{X})$  by taking partial derivatives of the right-hand sides of the equations of motion with respect to each component of  $\vec{X}$ . The system's structure dictates that the Jacobian has a canonical block form:

$$A = \begin{bmatrix} \mathbf{0}_{3 \times 3} & I_{3 \times 3} \\ D^2\Omega & C \end{bmatrix} \quad (54)$$

where:

- $I_{3 \times 3}$  is the identity matrix coming from the time derivatives of the position variables,
- $D^2\Omega$  is the Hessian of the effective potential  $\Omega$  with respect to  $x, y, z$ ,
- $C$  contains the Coriolis coupling:  $C_{4,5} = 2$ ,  $C_{5,4} = -2$ .

We compute the eigenvalues  $\lambda_i$  and eigenvectors  $\vec{v}_i$  of the Jacobian  $A$ . Their physical interpretation is:

- $\Re(\lambda_i) > 0$ : unstable direction  $\rightarrow$  manifold  $\mathcal{W}^u$
- $\Re(\lambda_i) < 0$ : stable direction  $\rightarrow$  manifold  $\mathcal{W}^s$
- $\lambda_i = \pm i\omega$ : oscillatory (center)  $\rightarrow$  periodic orbits around equilibrium

Complex eigenvalues yield real dynamics via:

$$\vec{x}(t) = e^{\sigma t} (\Re(\vec{v}) \cos(\omega t) + \Im(\vec{v}) \sin(\omega t)) \quad (55)$$

We construct real bases:

$$\vec{v}_1 = \Re(\vec{v}), \quad \vec{v}_2 = \Im(\vec{v}) \quad (56)$$

and use linear combinations to initialize oscillatory trajectories. To approximate the invariant manifolds:

$$\vec{x}_0 = \vec{x}_{SL1} + \varepsilon \cdot \hat{v}_u \quad (\text{unstable}) \quad (57)$$

$$\vec{x}_0 = \vec{x}_{SL1} + \varepsilon \cdot \hat{v}_s \quad (\text{stable}) \quad (58)$$

We then integrate the system forward (for  $\mathcal{W}^u$ ) or backward (for  $\mathcal{W}^s$ ) in time.

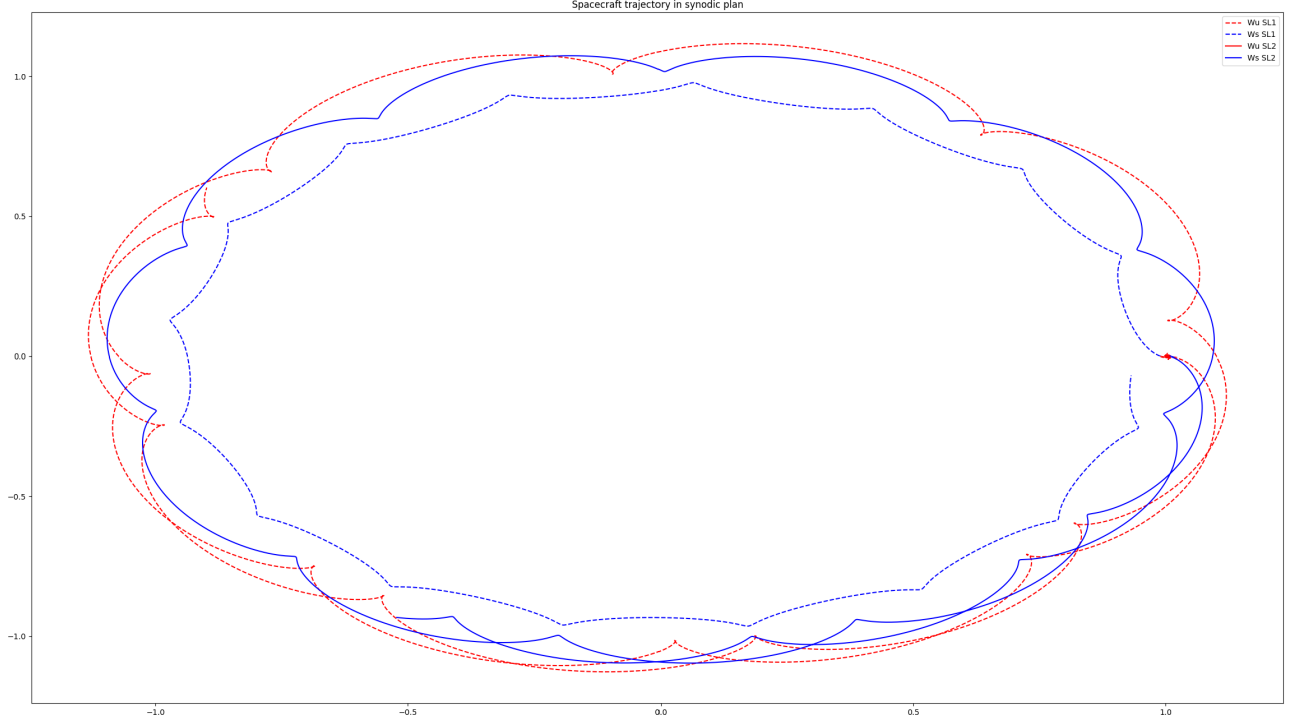


Figure 27: Spacecraft trajectory from  $L_1$  and  $L_2$  with solar sails

The figure 27 illustrates the manifolds generated from the  $L_1$  and  $L_2$  points when the solar sail acceleration is included. The plotted trajectories show the natural flow of particles initially displaced along the stable and unstable eigenvectors of the linearized system. The characteristic spiral-like structures are consistent with the theoretical manifolds presented in [Farrés et al., 2019], confirming the correct implementation of the sail-augmented CRTBP model. This preliminary work constituted the first stage of the internship and served primarily as a training exercise to develop familiarity with the CRTBP and the computation of invariant manifolds.